

Theory of Clamped-Clamped Bistable Plate

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1 Introduction

The actuation of mechanical metamaterials has historically relied on direct physical intervention, such as localized point forces, contact with actuators (pneumatic, fluidic, electro-magnetic), or the embedding of rigid boundary networks, as mainly seen in the previous sections. However, actuation does not necessarily require solid boundaries or embedded electronics; it can also be harvested directly from ambient environmental fields. Among these, fluid flow represents a medium that is continuously interacting with structures (FSI) giving rise to a closed-loop interaction that can be harnessed for passive control, energy harvesting, or adaptive response. The integration of mechanical metamaterials with fluid dynamics (Fluid Metamaterial Interaction - FMI) opens entirely new paradigms for passive actuation, adaptive aerodynamics, and self-regulating structures.

Mechanical metamaterials, particularly phononic crystals and elastic/acoustic metamaterials, have already been investigated in flow environments for their ability to exhibit tailored dynamic responses when excited at specific frequencies. A foundational contribution was introduced by Hussein et al. (2015) [?], who embedded periodic elastic metamaterials beneath a smooth surface to interfere destructively with Tollmien–Schlichting (TS) waves in a boundary layer. By designing the subsurface dispersion relation to produce out-of-phase surface motion, they passively delayed laminar-to-turbulent transition without altering the outer geometry. Subsequent extensions by Kianfar et al. [?] demonstrated subwavelength operation using locally resonant elastic metamaterials, achieving the same stabilization at frequencies far below the Bragg limit and therefore reducing the required subsurface thickness. Building on these ideas, Keogh et al. [?] experimentally realized an ultra-thin phononic metamaterial subsurface composed of layered spirals and voids; they showed attenuation of targeted vortex frequencies in a turbulent boundary layer while preserving a smooth outer surface. Complementary numerical studies, such as Ramakrishnan et al. [?], have examined phononic surfaces in turbulent channel flow, confirming that near-wall coherent structures can be disrupted when the metamaterial’s spatio-temporal scales match those of the flow. A recent comprehensive review by Avallone et al. [?] synthesizes these efforts, highlighting the applications of metamaterials

ranging from drag reduction on maritime hulls and aircraft wings to noise suppression in ducts and vibration isolation in marine structures.

In nearly all of these studies, the metamaterial displacements remain small (linear regime), so the flow perceives an essentially fixed geometry and the back-action of shape change on the pressure field is negligible. While a handful of investigations have begun to explore this two-way coupling, the literature lacks systematic treatment of large geometric reconfigurations and the resulting nonlinear pressure feedback. Reprogrammable mechanical metamaterials, which incorporate bistable or multistable unit cells, inherently produce large displacements and snap-through instabilities. These features open a new regime of FSI in which the metamaterial shape and the flow pressure profile co-evolve in a highly nonlinear loop. Understanding how incoming flows trigger (or suppress) snap-through, and how the rapid reconfiguration driven by the instability in turn alters vortex shedding, boundary-layer stability, etc., is therefore essential for the rational design of multistable metamaterials that can adapt dynamically to their fluid environment and leverage their nonlinear and high-amplitude strain field. A growing body of work has begun to address exactly this coupling. Kim et al. [?] performed wind-tunnel experiments and low-order simulations on a buckled elastic sheet under uniform external flow, identifying the critical velocity for triggering the instability and the subsequent periodic snap-through oscillations. Mahravan et al. [?] extended this to wall-bounded configurations, demonstrating how confinement lowers the critical velocity and produces distinct contact modes (sliding, bouncing) that annihilate vortices and modify the wake.

For true bistable architectures, Gomez et al. [?] derived an analytical lubrication-theory model of a post-buckled arch along a channel sidewall. This approach was later generalized by Bonthron and Tubaldi [?] with generalized torsional springs at the boundaries. By varying the support stiffness they spanned the full range from clamped-clamped to hinged-hinged, mapping critical flow rates for snap-through and showing that boundary compliance provides an additional design knob for passive flow-rate regulation. Ma et al. [?] performed high-fidelity simulations of bistable beams in uniform flow, comparing bistable and monostable responses and revealing how snap-through releases stored strain energy faster than it accumulates, with implications for drag reduction and energy harvesting. Complementary studies on inclined-clamp sheets (Mahravan et al. [?]) further illustrate how geometric asymmetry modulate the snap-through thresholds and post-critical oscillation modes. These investigations establish that multistable structures, with snap-through dynamics, are ideally suited for passive fluidic functionalities like flow control via adaptive valves that snap at prescribed flux rates, turbulence reduction by rapid reconfiguration that disrupts coherent structures, vortex generation for enhanced mixing or heat transfer in dual-channel systems, and broadband energy harvesting by converting periodic snap-through kinetic energy into electrical output.

Researchers have also examined the response of thin plates in axial inviscid flow using linear potential-flow theory. These studies traces back to the works of Guo and Paidoussis [?, ?]. In these studies the authors examined the hy-

droelastic stability of thin rectangular plates immersed in a two-dimensional inviscid channel flow using linear potential-flow theory. For a single plate with free side-edges, the perturbation potential satisfies Laplace’s equation with impermeability conditions on the plate and channel walls; the plate deflection is discretized via a Galerkin expansion. The resulting eigenvalue problem yields the complex eigenfrequencies as functions of flow velocity, revealing that divergence (static loss of stability) and coupled-mode flutter can occur for a wide range of boundary conditions, while single-mode flutter is possible only for non-symmetric supports. Extending the framework to multiple plates, they showed that increasing plate length or decreasing channel height lowers both natural frequencies and critical velocities. These works provided a systematic mapping of stability boundaries for plates in confined axial flow, establishing the mechanisms of aerodynamic softening, where the convective term generates a negative fluid-stiffness contribution proportional to U^2 , that monotonically reduces modal frequencies until the lowest eigenvalue reaches zero (divergence).

Although originally derived for liquid flows, the same potential-flow Galerkin methodology was rapidly adopted for air flows. Eloy et al. [?] applied an analogous reduced-order model to cantilevered flexible plates in uniform air flow, combining linear potential theory with experimental wind-tunnel tests to quantify the effect of aspect ratio on the flutter threshold. Subsequent theoretical and experimental studies, such as Gibbs [?] on rectangular plates with fixed leading edges in three-dimensional axial air flow, retained the same potential flow approach while incorporating structural nonlinearity or vortex-lattice corrections for higher accuracy in air. These air-flow extensions demonstrated that the fundamental softening and divergence mechanisms identified in the liquid case persist in compressible subsonic regimes.

Tubaldi and Amabili [?] considered an infinitely wide rectangular plate, periodically supported in both directions and immersed in a confined axial flow on one side. The same reduced-order framework has been extended to imperfect and geometrically nonlinear plates.

Such potential-flow Galerkin models are deliberately low-fidelity: they neglect viscosity, separation, and vortex shedding. Their strength, however, lies in the fact that the matrices are assembled analytically, the eigenvalue problem can be solved in seconds for hundreds of parameter combinations (aspect ratio, channel height, pre-compression, imperfection amplitude), delivering clear stability boundaries and scaling laws that would be prohibitively expensive to obtain with high-fidelity simulations.

The analytical approach is particularly valuable for the design of multi-stable mechanical metamaterials. High-fidelity viscous FSI simulations become prohibitively expensive in this regime: snap-through events involve large strains, geometric nonlinearity, even more when hundreds of such bistable units are arranged in a periodic lattice, with propagating transition waves, tunable bandgaps, or adaptive shape reconfiguration, with strong two-way coupling between the deforming lattice and the surrounding flow.

In contrast, the linear potential-flow reduced-order framework researchers can extract simple scaling rules, like how the critical snap-through velocity scales

with the prestrain or how the natural frequencies of the two wells shift as the freestream speed increases. These scaling laws allow rapid exploration of the design space before any high-fidelity simulation is performed. In particular, the competition between aerodynamic softening and flow-induced membrane tension can be quantified explicitly, revealing parameter windows where the net response is hardening, softening, or neutrally stable.

Recently we demonstrated a system, comprised of a plate resting on foundations of symmetric bistable elements [?], supporting transition waves which was designed to interact with flow via the continuous plate surface, as opposed to the generally porous structure of classic metamaterials. The ability to predict how an oncoming flow modifies the local bistable potential of each cell makes it possible to engineer lattices in which transition waves are triggered, moved, or suppressed by the external flow field. In short, the potential-flow approach does not replace high-fidelity solvers; it provides an analytical roadmap for the design of flow-interactive multistable metamaterials.

In this work we close the remaining modeling gap by studying the FSI problem of a clamped–clamped bistable plate immersed on one side in flow. The structural model employs a Galerkin reduction, capturing asymmetric double-well potentials, and snap-through bifurcation. The fluid domain is treated as potential-flow and discretized by a boundary-integral (of discrete sources), yielding an unsteady Bernoulli pressure. The modal frequencies are calculated as function of prestrain and flow velocity. Finally, the nonlinear dynamics regime is studied by introducing discrete vortices as representation of a near-wall KVS. These can excite the plate to snap-through with various regimes.

2 Beam Model

We model the beam as clamped-clamped, of length L , cross-sectional area A , moment of inertia I , Young's modulus E , mass per unit length $m = \rho A$, under axial end shortening $\Delta > \Delta_{cr}$ leading to buckling.

2.1 Dimensional Energies

The dimensional strain energy U consists of bending and membrane contributions:

$$U = \frac{1}{2}EI \int_0^L (w''_{tot})^2 dx + \frac{1}{2}EAL\epsilon_0^2, \quad (1)$$

where $w_{tot}(x, t) = w(x, t) + w_0(x)$ is the transverse deflection from the flat position, $w(x, t)$ is the transverse deflection from the slightly deformed position $w_0(x)$ (the initial imperfection), and ϵ_0 is the constant midplane strain given by

$$\epsilon_0 = \frac{1}{L} \left(\frac{1}{2} \int_0^L ((w'_{tot})^2 - (w'_0)^2) dx - \Delta \right) = \frac{1}{L} \left(\frac{1}{2} \int_0^L (w')^2 dx + \int_0^L w' w'_0 dx - \Delta \right). \quad (2)$$

The dimensional kinetic energy T is

$$T = \frac{1}{2}m \int_0^L \dot{w}^2 dx. \quad (3)$$

2.2 Non-Dimensionalization

Introduce non-dimensional variables: $\zeta = x/L$, $W = w/L$, $\tau = t\sqrt{EI/(mL^4)}$, $\bar{\Delta} = \Delta/L$.

Then, $w = LW$, $w' = W'$, $w'' = W''/L$, $\dot{w} = L\sqrt{EI/(mL^4)}\bar{W}$, where $\bar{\tau} = d/d\tau$.

The integral transforms as

$$\int_0^L (w')^2 dx = L \int_0^1 (W')^2 d\zeta. \quad (4)$$

Thus,

$$\epsilon_0 = -\bar{\Delta} + \frac{1}{2} \int_0^1 (W')^2 d\zeta + \frac{1}{2} \int_0^1 W' W'_0 d\zeta. \quad (5)$$

The bending energy becomes

$$U_{\text{bending}} = \frac{1}{2}EI \frac{1}{L} \int_0^1 (W'')^2 d\zeta. \quad (6)$$

The membrane energy is

$$U_{\text{membrane}} = \frac{1}{2}EAL \left(-\bar{\Delta} + \frac{1}{2} \int_0^1 (W')^2 d\zeta + \int_0^1 W' W'_0 d\zeta \right)^2. \quad (7)$$

To derive T : $\dot{w}^2 = L^2(EI/(mL^4))\bar{W}^2$, so

$$\int_0^L \dot{w}^2 dx = L^3 \frac{EI}{mL^4} \int_0^1 \bar{W}^2 d\zeta, \quad (8)$$

and multiplying by $\frac{1}{2}m$ gives the expression:

$$T = \frac{1}{2} \frac{EI}{L} \int_0^1 \bar{W}^2 d\zeta. \quad (9)$$

The dimensional energies can be re-written as $U = \frac{EI}{L} \bar{U}$, where $\bar{U}$ is the non dimensional energy.

2.3 Galerkin Method

Assume $W(\zeta, \tau) = \sum_{i=1}^N q_i(\tau) \phi_i(\zeta)$, where ϕ_i are clamped-clamped beam mode shapes satisfying $\phi_i(0) = \phi_i(1) = 0$, $\phi_i'(0) = \phi_i'(1) = 0$.

2.3.1 Clamped-Clamped Beam Mode Shapes

The spatial mode shapes $\phi(x)$ of a uniform beam are governed by the Euler-Bernoulli equation:

$$\phi''''(\zeta) - \lambda^4 \phi(\zeta) = 0. \quad (10)$$

The general solution to this ordinary differential equation is a linear combination of trigonometric and hyperbolic functions:

$$\phi(\zeta) = C_1 \cosh(\lambda\zeta) + C_2 \cos(\lambda\zeta) + C_3 \sinh(\lambda\zeta) + C_4 \sin(\lambda\zeta) \quad (11)$$

For a clamped-clamped plate the boundary conditions are:

$$\phi(0) = \phi'(0) = 0, \quad \text{and} \quad \phi(1) = \phi'(1) = 0 \quad (12)$$

Applying these at $\zeta = 0$ yields $C_2 = -C_1$ and $C_4 = -C_3$. Substituting into the general solution at $\zeta = 1$ gives the system:

$$\begin{bmatrix} \cosh(\lambda) - \cos(\lambda) & \sinh(\lambda) - \sin(\lambda) \\ \sinh(\lambda) + \sin(\lambda) & \cosh(\lambda) - \cos(\lambda) \end{bmatrix} \begin{Bmatrix} C_1 \\ C_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (13)$$

For a non-trivial solution to exist, the determinant of this matrix must be zero. Computing the determinant and simplifying using hyperbolic and trigonometric identities ($\cosh^2 \lambda - \sinh^2 \lambda = 1$ and $\cos^2 \lambda + \sin^2 \lambda = 1$) yields the equation:

$$\cosh(\lambda) \cos(\lambda) - 1 = 0 \quad (14)$$

The values of λ_i for each mode $i \in [1, N]$ are found by numerically solving this equation. Once λ_i is known, the ratio of the coefficients $\sigma_i = -C_3/C_1$ can be extracted from the first row of the matrix:

$$\sigma_i = \frac{\cosh(\lambda_i) - \cos(\lambda_i)}{\sinh(\lambda_i) - \sin(\lambda_i)} \quad (15)$$

Setting $C_1 = 1$, the analytical expression for the i -th unnormalized mode shape becomes:

$$\tilde{\phi}_i(x) = \cosh(\lambda_i \zeta) - \cos(\lambda_i \zeta) - \sigma_i \left(\sinh(\lambda_i \zeta) - \sin(\lambda_i \zeta) \right) \quad (16)$$

Finally, to ensure the Galerkin decomposition gives an orthonormal basis, the mode shapes are normalized as:

$$\phi_i(x) = \frac{\tilde{\phi}_i(x)}{\sqrt{\int_0^1 \tilde{\phi}_i(x)^2 dx}} \quad (17)$$

2.3.2 Energies using Galerkin decomposition

Kinetic term:

$$\bar{T} = \frac{1}{2} \sum_i \dot{q}_i^2. \quad (18)$$

Bending term:

$$\bar{U}_{bending} = \frac{1}{2} \sum_i \lambda_i^4 q_i^2, \quad (19)$$

since $\int_0^1 (\phi_i'')(\phi_j'') d\zeta = \lambda_i^4 \delta_{ij}$.

Membrane term:

$$\bar{U}_{membrane} = \frac{1}{2} \alpha \left(-\bar{\Delta} + \frac{1}{2} \sum_{i,j} q_i q_j B_{ij} + \sum_{i,j} q_i q_{0j} B_{ij} \right)^2, \quad (20)$$

where $\alpha = EAL^2/EI = L^2/r^2$, and $r^2 = I/A$, q_0 is the modal coefficient of the initial imperfection, and where $B_{ij} = \int_0^1 (\phi_i')(\phi_j') d\zeta$. For orthogonal modes, $B_{ij} = \delta_{ij} B_{ii}$, with $B_{ii} = \int_0^1 (\phi_i')^2 d\zeta \approx 12.36$ for the first mode.

The total structural potential energy is:

$$\bar{U}_{struct}(\mathbf{q}) = \frac{1}{2} \sum_{i=1}^N \lambda_i^4 q_i^2 + \frac{1}{2} \alpha \left(-\bar{\Delta} + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N B_{ij} q_i q_j + \sum_{i=1}^N \sum_{j=1}^N B_{ij} q_i q_{0j} \right)^2. \quad (21)$$

2.4 Euler-Lagrange Equations

The Euler-Lagrange equation for each q_k is

$$\frac{d}{d\tau} \left(\frac{\partial \bar{\mathcal{L}}}{\partial \dot{q}_k} \right) - \frac{\partial \bar{\mathcal{L}}}{\partial q_k} = 0. \quad (22)$$

We have $\partial \bar{\mathcal{L}} / \partial \dot{q}_k = \dot{q}_k$, so $d/d\tau (\partial \bar{\mathcal{L}} / \partial \dot{q}_k) = \ddot{q}_k$.

Let $S = -\bar{\Delta} + \frac{1}{2} \sum_{i,j} B_{ij} q_i q_j + \sum_{i,j} B_{ij} q_i q_{0j}$. Then,

$$\frac{\partial S}{\partial q_k} = \sum_i q_i B_{ik} + \sum_i q_{0i} = \sum_i (q_i + q_{0i}) B_{ik}. \quad (23)$$

Thus,

$$\frac{\partial \bar{\mathcal{L}}}{\partial q_k} = -\lambda_k^4 q_k - \alpha S \frac{\partial S}{\partial q_k}. \quad (24)$$

The unforced equation of motion is

$$\ddot{q}_k + \lambda_k^4 q_k + \alpha \left(-\bar{\Delta} + \frac{1}{2} \sum_{i,j} B_{ij} q_i q_j + \sum_{i,j} B_{ij} q_i q_{0j} \right) \sum_i (q_i + q_{0i}) B_{ik} = 0. \quad (25)$$

3 Point force results

3.1 Static validation of mechanical model

Fig. 1 compares the force–displacement response predicted by the reduced-order Galerkin beam model against an Abaqus finite-element (FEA) simulation, demonstrating agreement between the two. The FEA model is built as a 2D planar shell representation of the plate with an assigned out-of-plane thickness equal to the physical depth, meshed with structured quadrilateral CPS4I elements; both ends are constrained through kinematically coupled reference points to enforce end shortening. The loading sequence is designed as follow: (i) a nonlinear static step applies the prescribed pre-compression, aided by a small transverse perturbation force at the actuation location to select the buckled configuration, (ii) the perturbation is removed, and (iii) an implicit-dynamics step with moderate numerical dissipation imposes a ramped transverse displacement at the actuation point to drive snap-through while recording the reaction force and displacement history. The close overlap between the Abaqus curve and the Galerkin predictions indicates that the reduced model captures the key nonlinear mechanics relevant to bistable structures—geometric nonlinearity, and the loss of stability event associated with snap-through at a fraction of the computational cost.

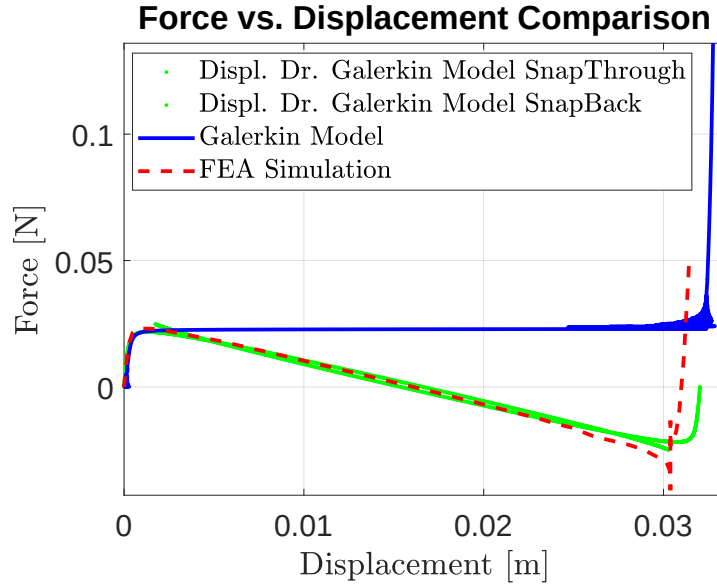


Figure 1: Validation of the mechanical model based on Galerkin decomposition

3.2 Point Force stability analysis

In Fig. 2 we present the results for the stability analysis when increasing imperfections, defined in the previous subsection, are introduced. In the first plot (a) the critical point load F_{crit} (located at $0.5L$) required to snap down the plate is shown. As this plot shows, F_{crit} decays non-linearly when ϵ is increased.

Plot (b) shows the evolution of the minimum eigenvalue, which determines the stiffness of the softest mode in the system. The force at which this reaches 0, marks the onset of instability, and also determines, but looking at the associated eigenvector, what's the unstable shape.

In plot (b), we show the projection of the eigenvector associated with the minimum eigenvalue, onto the equilibrium position. For a perfect system ($\epsilon = 0\%$) the equilibrium shape is orthogonal to the preferred snapping direction (see Fig. 3(c-e)). When the imperfection is increased, but still small ($\epsilon < 20\%$), we can notice how the projection value increases monotonically until the onset of instability: the plate softest mode, as shown in Fig. 3(d), is a mix of mode 1 and mode 2, while the equilibrium position (Fig. 3(f)) is mostly governed by mode 1. When the applied force is increased, the asymmetry of the equilibrium configuration (governed mainly by mode 2) increases, therefore increasing the projection value. When higher values of imperfections ($\epsilon > 20\%$) are used, we can see how the projection initially increases value, but then suddenly decreases it before snap-through. This because the eigenvector associated with the minimum eigenvalue has a stronger component of mode 1, which decreases in the equilibrium shape for higher loads, therefore marking a decrease in the projection value.

Overall, as expected, the interaction between the applied symmetric point force and the antisymmetric Mode 2 imperfection triggers a symmetry-breaking bifurcation earlier than in the perfect geometry.

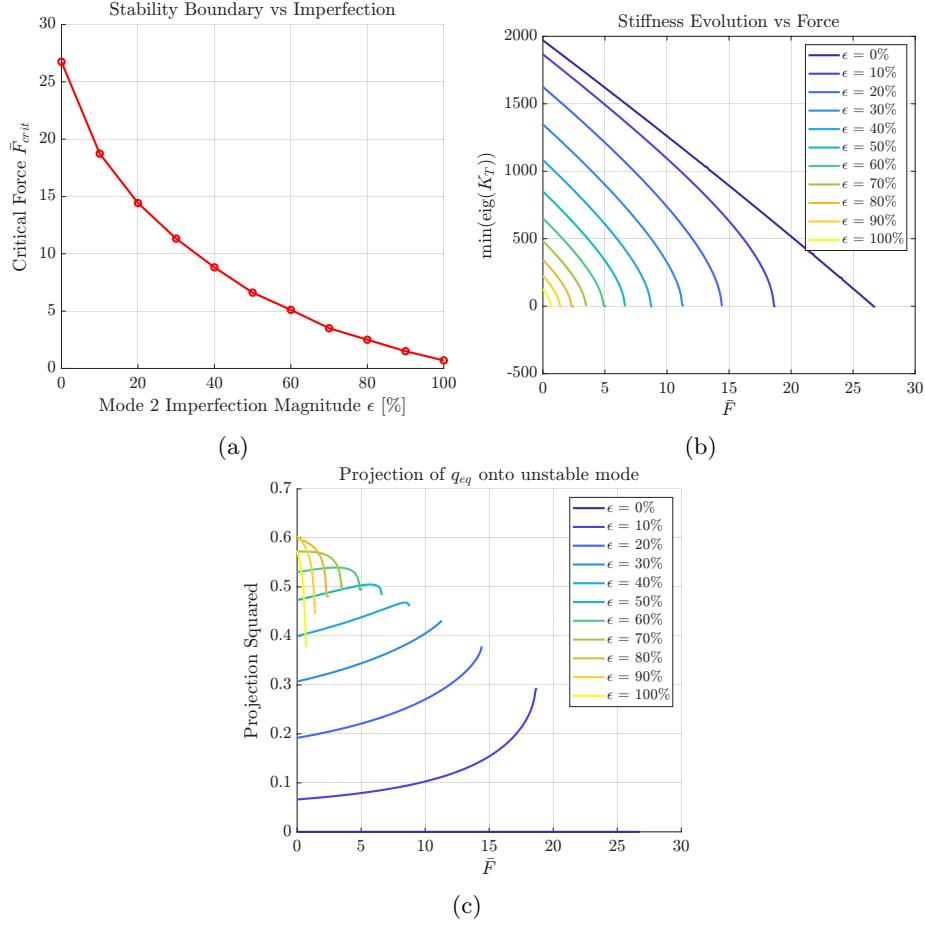


Figure 2: Stability analysis of the bistable plate subject to a transverse point force under varying mode 2 imperfection magnitudes (ϵ). (a) The critical snap-through force $\bar{F}_{crit}$ monotonically decreases as the geometric imperfection increases. (b) Evolution of the minimum eigenvalue of the tangent stiffness matrix against applied force, marking the onset of instability when the stiffness reaches zero. (c) Squared projection of the equilibrium shape q_{eq} onto the unstable mode vector. For perfect systems ($\epsilon = 0\%$), the initial projection is near zero but grows due to symmetry-breaking; for highly imperfect systems ($\epsilon > 20\%$), the projection initially increases but drops prior to snap-through.

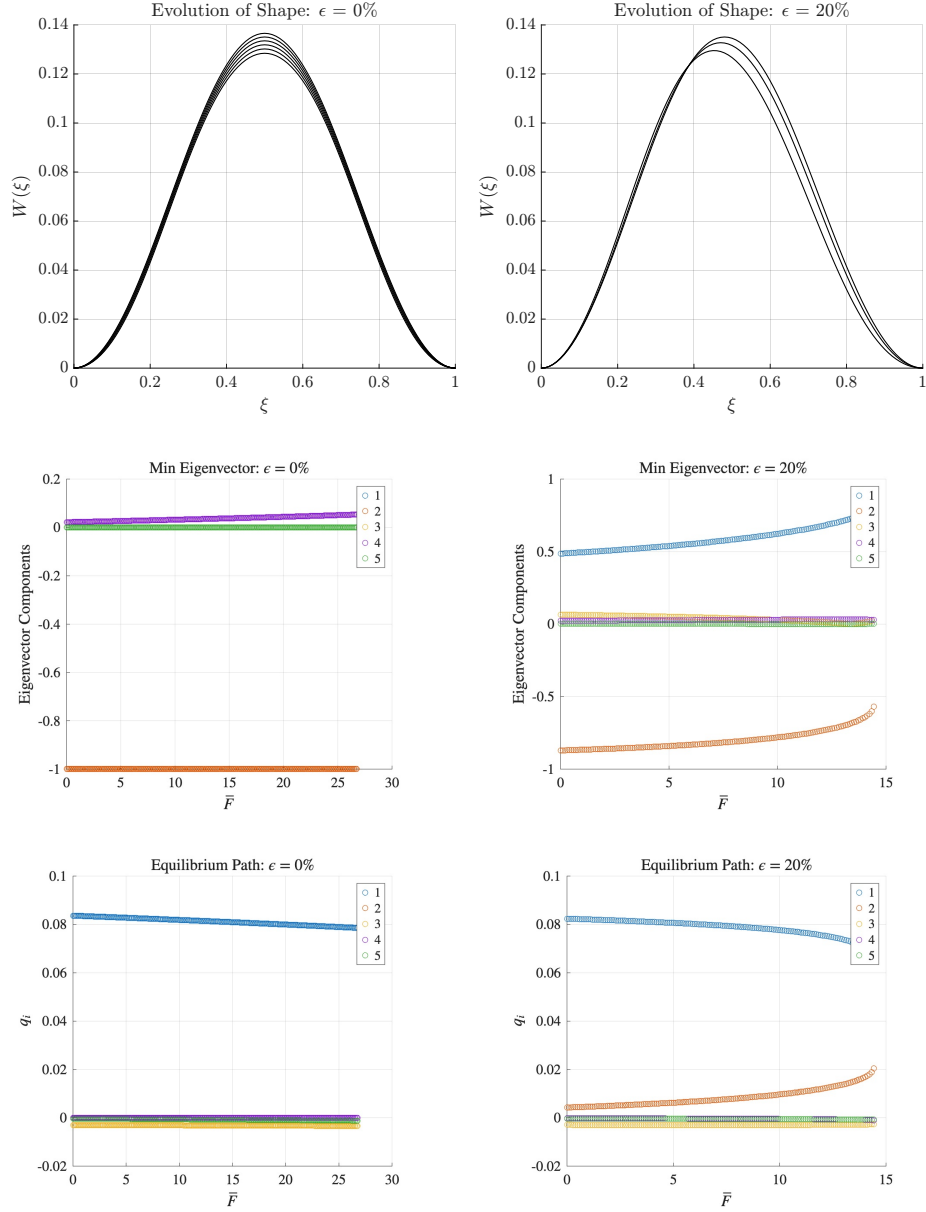


Figure 3: Pre-buckling shape evolution and linear stability characteristics for a perfect plate ($\epsilon = 0\%$, left column) and an imperfect plate ($\epsilon = 20\%$, right column). (a, b) Transverse deflection profile $W(\xi)$ showing increased asymmetry for the imperfect case under load. (c, d) Evolution of eigenvector components associated with the minimum eigenvalue, dictating the shape of the unstable mode. (e, f) Trajectory of the equilibrium coordinates q_k as the applied force $\bar{F}$ increases towards the critical snap-through threshold, highlighting the coupled activation of antisymmetric modes in the imperfect structure.

4 Dynamics

4.1 Small amplitude oscillations

To study the linear stability and vibrations of the plate around its buckled equilibrium, we first have to analyze the properties of the nonlinear coupling matrix B and the tangent stiffness of the system.

The system dynamics are governed by the interaction coefficients defined as:

$$B_{ij} = \int_0^1 \phi'_i(\zeta) \phi'_j(\zeta) d\zeta \quad (26)$$

Numerical evaluation of this matrix reveals a sparse structure where specific off-diagonal terms vanish ($B_{ij} = 0$). This sparsity is not accidental but stems from the parity of the mode shapes for a clamped-clamped beam.

- **Symmetric Modes** ($k = 1, 3, 5, \dots$): These modes are even functions with respect to the midpoint ($\zeta = 0.5$). Their derivatives ϕ'_k are therefore odd (antisymmetric) functions.
- **Antisymmetric Modes** ($k = 2, 4, 6, \dots$): These modes are odd functions with respect to the midpoint. Their derivatives ϕ'_k are therefore even (symmetric) functions.

The coefficient B_{ij} represents the inner product of these derivatives. If indices i and j have different parity (one even, one odd), the integrand $\phi'_i \phi'_j$ is the product of a symmetric and an antisymmetric function, resulting in an antisymmetric function. Integrating an antisymmetric function over a symmetric domain yields zero. Consequently, we observe two distinct interaction behaviors relative to the fundamental buckling mode ($k = 1$):

1. **Decoupled Antisymmetric Modes:** For even k (Modes 2, 4), $B_{1k} = 0$. These modes are geometrically decoupled from the fundamental mode in the linearized regime.
2. **Coupled Symmetric Modes:** For odd $k \neq 1$ (Modes 3, 5), $B_{1k} \neq 0$. These modes share the same symmetry as the fundamental mode, resulting in a non-zero metric coupling.

4.1.1 Linearization of the Equation of Motion

To study the linear stability and vibrations of the plate around its buckled equilibrium, we perform a linearization of the full Equation of Motion. We consider small time-dependent perturbations $\delta q_k(\tau)$ around the static equilibrium vector q_{eq} . Assuming the beam is buckled purely in the first mode, the equilibrium vector is defined as $q_{eq} = \{q_1, 0, \dots, 0\}^T$. The generalized coordinate for the k -th mode is:

$$q_k(\tau) = q_{eq,k} + \delta q_k(\tau) \quad (27)$$

We begin with the full non-linear Equation of Motion for mode k :

$$\ddot{q}_k + \lambda_k^4 q_k + \alpha \left[-\bar{\Delta} + \frac{1}{2} \sum_{m,n} q_m q_n B_{mn} \right] \sum_i q_i B_{ik} = 0 \quad (28)$$

Substituting the perturbed coordinate $q = q_{eq} + \delta q$ into the equation:

$$\delta \ddot{q}_k + \lambda_k^4 (q_{eq,k} + \delta q_k) + \alpha \left[-\bar{\Delta} + \frac{1}{2} \sum_{m,n} (q_{eq,m} + \delta q_m)(q_{eq,n} + \delta q_n) B_{mn} \right] \sum_i (q_{eq,i} + \delta q_i) B_{ik} = 0 \quad (29)$$

We expand the non-linear terms, retaining only components up to the first order in δq . The tension-like term (in brackets) expands as:

$$[\dots] \approx \underbrace{-\bar{\Delta} + \frac{1}{2} \sum_{m,n} q_{eq,m} q_{eq,n} B_{mn}}_{S_{eq} \text{ (Static Tension)}} + \underbrace{\sum_{m,n} q_{eq,m} \delta q_n B_{mn}}_{\text{Dynamic Perturbation}} \quad (30)$$

Utilizing the unimodal equilibrium assumption ($q_{eq,m} = 0$ for $m \neq 1$), the dynamic perturbation simplifies to $q_1 \sum_n \delta q_n B_{1n}$. Similarly, the geometric projection term expands to:

$$\sum_i (q_{eq,i} + \delta q_i) B_{ik} = q_1 B_{1k} + \sum_i \delta q_i B_{ik} \quad (31)$$

We now multiply the expanded bracket by the expanded sum, neglecting second-order products ($\delta q \cdot \delta q$). The interaction term becomes:

$$\alpha \left[S_{eq} (q_1 B_{1k}) + S_{eq} \sum_i \delta q_i B_{ik} + \left(q_1 \sum_n \delta q_n B_{1n} \right) (q_1 B_{1k}) \right] \quad (32)$$

Substituting this back into the Equation of Motion and cancelling the static equilibrium terms ($\lambda_k^4 q_{eq,k} + \alpha S_{eq} q_1 B_{1k} = 0$), we obtain the linearized equation for the perturbation δq_k :

$$\delta \ddot{q}_k + \lambda_k^4 \delta q_k + \alpha S_{eq} \sum_i \delta q_i B_{ik} + \alpha q_1^2 B_{1k} \sum_n \delta q_n B_{1n} = 0 \quad (33)$$

To determine the natural frequency of mode k , we isolate the diagonal stiffness term $(K_T)_{kk}$ by collecting the coefficients of δq_k (setting $i = k$ and $n = k$ in the summations):

$$\omega_k^2 = (K_T)_{kk} = \lambda_k^4 + \underbrace{\alpha S_{eq} B_{kk}}_{\text{Stress Softening}} + \underbrace{\alpha q_1^2 B_{1k}^2}_{\text{Geometric Stiffening}} \quad (34)$$

4.1.2 Modal Frequencies

From the equilibrium condition of Mode 1, the general stiffness equation yields three distinct cases:

Case 1: The Fundamental Mode ($k = 1$) Substituting $k = 1$, the linear stiffness cancels exactly with the static tension term ($\lambda_1^4 - \lambda_1^4 = 0$). The restoring force is purely nonlinear:

$$\omega_1^2 = \alpha B_{11}^2 q_1^2 \quad (35)$$

Case 2: Antisymmetric Higher Modes ($k = 2, 4, \dots$) Since $B_{1k} = 0$, the geometric stiffening term vanishes. The frequency is determined solely by the saturated static tension:

$$\omega_k^2 = \lambda_k^4 - \lambda_1^4 \frac{B_{kk}}{B_{11}} \quad (36)$$

These frequencies remain constant in the post-buckling regime.

Case 3: Symmetric Higher Modes ($k = 3, 5, \dots$) Since $B_{1k} \neq 0$, these modes retain the geometric stiffening term:

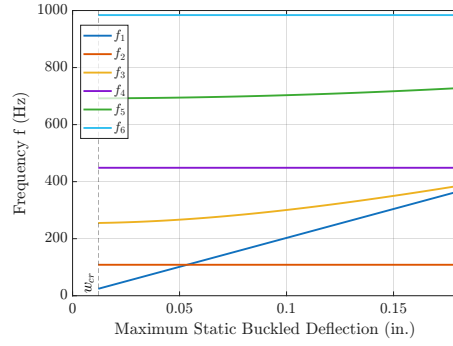
$$\omega_k^2 = \left(\lambda_k^4 - \lambda_1^4 \frac{B_{kk}}{B_{11}} \right) + \alpha B_{1k}^2 q_1^2 \quad (37)$$

These modes exhibit weak stiffening proportional to the square of the buckling amplitude.

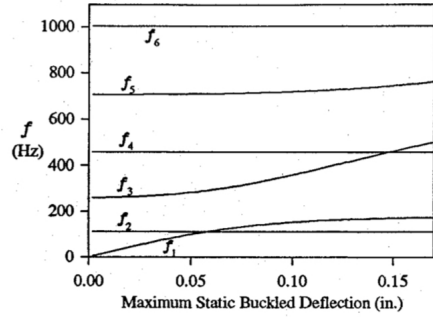
These analytical results, however, are valid only for a small subset of parameters: as can be seen in Fig. 4, which shows the comparison with [?] of each mode's frequencies for different buckled deformation (which is characteristic of the imposed prestrain Δ), the frequencies match only for very small buckled deflections/prestrain. Fig. 5, instead, shows the comparison, and match, obtained by solving the static equilibrium equation to find q_{eq} . Solving then the eigen problem $(\mathbf{K}_T - \omega^2 \mathbf{I})\mathbf{v} = \mathbf{0}$ gives the frequencies associated with each eigenvector.

As illustrated in Fig. ??, the modal frequencies exhibit distinct behaviors based on their symmetry. As the pre-strain $\bar{\Delta}$ is increased, the fundamental frequency f_1 drops to zero at the critical buckling threshold $\bar{\Delta}_{cr}$, marking the onset of static divergence. In the post-buckling regime ($\bar{\Delta} > \bar{\Delta}_{cr}$), the fundamental mode and higher-order symmetric modes (e.g., ϕ_3, ϕ_5). Conversely, the antisymmetric modes (ϕ_2, ϕ_4) decouple geometrically from the primary buckling mode, resulting in natural frequencies that remain relatively constant across the post-buckled parameter space.

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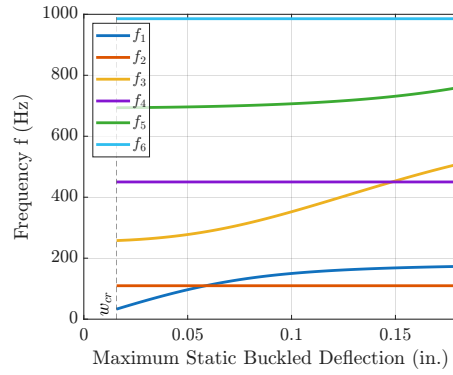


(a)

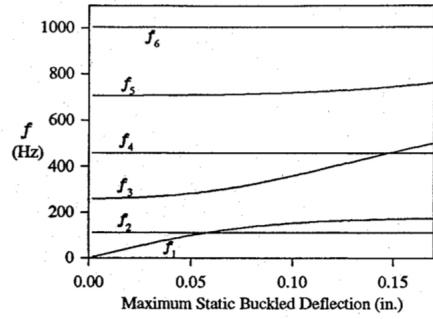


(b)

Figure 4: Comparison between the (a) analytical predictions and (b) [?].



(a)



(b)

Figure 5: Comparison between the (a) numerical predictions and (b) [?].

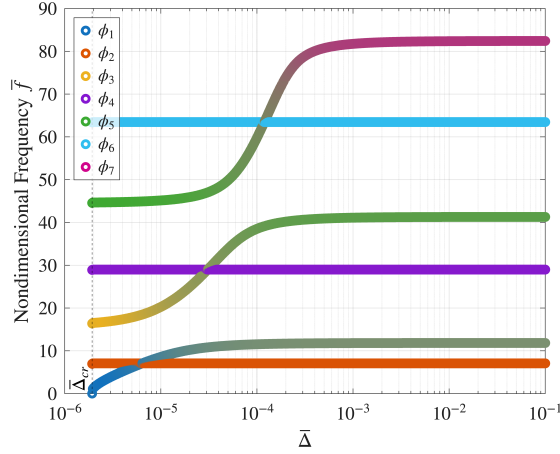


Figure 6: Evolution of the non-dimensional characteristic frequencies $\bar{f}_n$ for the Galerkin modes as a function of the static pre-strain $\bar{\Delta}$. The fundamental mode (ϕ_1) frequency drops to zero exactly at the critical buckling threshold $\bar{\Delta}_{cr}$. Symmetric modes show a stiffening behavior, while antisymmetric modes remain largely independent of the post-buckling amplitude. The color of each eigenmode changes for higher pre-strains, highlighting the shape change happening. The color is a mix dictated by the proportional presence of the fundamental modes in each eigenmode.

4.1.3 Nonlinear intrawell oscillations

Small-amplitude harmonic forcing reveals the intrawell resonance characteristics of the bistable plate, as shown in Fig. 7. When the external perturbation is applied exactly at the mid-span ($\xi = 0.50$, Fig. 7a), the forcing acts symmetrically, exclusively exciting the symmetric modes (ϕ_1, ϕ_3, ϕ_5). Consequently, distinct resonant peaks emerge strictly at the fundamental and higher-order symmetric natural frequencies. However, when the forcing location is shifted to break the spatial symmetry ($\xi = 0.45$, Fig. 7b), the energy distributes across the entire modal basis. This asymmetric excitation activates both symmetric and antisymmetric modes, giving rise to a denser spectrum of resonant peaks that align with the frequencies predicted by the linear stability analysis.

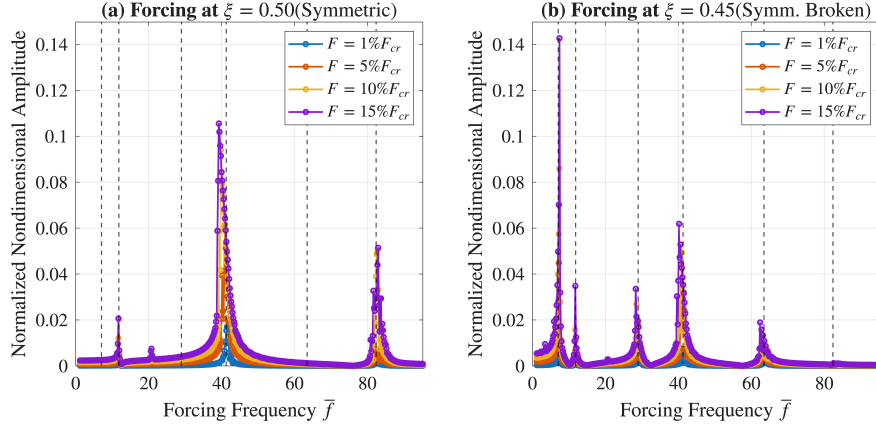


Figure 7: Response amplitudes for different frequencies $\bar{\Delta} = 2.5\%$.

4.1.4 Snap-through

Beyond small-amplitude vibrations, sufficiently high forcing amplitudes trigger the nonlinear snap-through instability, transferring the plate to its secondary equilibrium well. Fig. 8 presents the normalized response amplitude W/W_0 as a function of both forcing frequency and force amplitude (F/F_{crit}). The distinct "tongues" in the contour map indicate that the critical force required to induce snap-through is highly frequency-dependent. Specifically, the minimum force threshold necessary to dynamically trigger snapping occurs when the forcing frequency aligns with the plate's second and fourth lowest frequency resonance modes, which correspond to modes similar to mode $1 + 3$ and mode $3 + 5$.

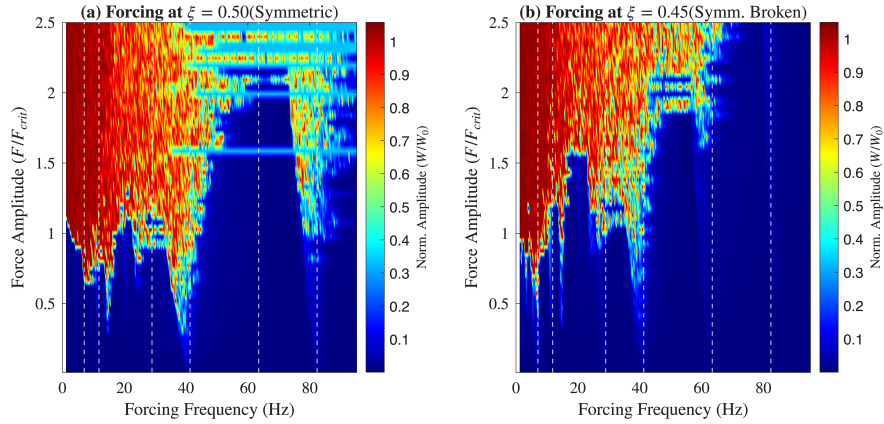


Figure 8: Response amplitudes for different frequencies $\bar{\Delta} = 2.5\%$.

5 Fluid Model

We now couple the clamped-clamped bistable plate to an inviscid, incompressible, irrotational fluid flowing over the plate. We assume a two-dimensional flow in the (X, Z) plane, with the undeformed plate lying along the X -axis and the fluid occupying the region above the plate. The plate centerline is at $Z = W(X, \tau)$, where the transverse displacement $W(X, \tau)$ is the same non-dimensional displacement used in the structural model.

5.1 Governing Equations and Velocity Potential

We consider a uniform incoming flow of speed U_∞ in the positive X -direction. We write the total velocity potential as

$$\Phi(X, Z, t) = U_\infty X + \phi(X, Z, t), \quad (38)$$

where:

- $U_\infty X$ is the potential of the uniform background flow expressed in terms of the non-dimensional coordinate X ;
- $\phi(X, Z, t)$ is the perturbation potential induced by the motion and deformation of the plate.

The fluid is incompressible and irrotational, so the perturbation potential satisfies Laplace's equation in the fluid domain:

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial X^2} + \frac{\partial^2 \phi}{\partial Z^2} = 0, \quad (39)$$

for all points (X, Z) in the fluid region above the plate, that is $Z > W(X, \tau)$. We also impose decay of perturbations far away from the plate:

$$\lim_{Z \rightarrow +\infty} \phi(X, Z, t) = 0. \quad (40)$$

The total fluid velocity field is given by

$$u(X, Z, t) = \frac{\partial \Phi}{\partial X} = \Phi_{,X} = U_\infty + \frac{\partial \phi}{\partial X}, \quad (41)$$

$$w_f(X, Z, t) = \frac{\partial \Phi}{\partial Z} = \Phi_{,Z} = \frac{\partial \phi}{\partial Z}, \quad (42)$$

where u is the streamwise velocity component and w_f is the normal (vertical) velocity component of the fluid.

5.2 No-penetration Condition

We parametrize the plate by the axial coordinate $X \in [0, 1]$, and the plate centerline is described by

$$\mathbf{r}_s(X, \tau) = (X, W(X, \tau)). \quad (43)$$

The tangent vector along the plate is

$$\mathbf{t}_0(X, \tau) = \frac{\partial \mathbf{r}_s}{\partial X} = (1, W_X(X, \tau)), \quad (44)$$

where we use the notation

$$W_X(X, \tau) = \frac{\partial W}{\partial X}(X, \tau). \quad (45)$$

The outward normal vector pointing into the fluid:

$$\mathbf{n}(X, \tau) = \frac{(-W_X(X, \tau), 1)}{\sqrt{1 + (W_X(X, \tau))^2}}. \quad (46)$$

We denote the components as (n_X, n_Z) . The plate velocity at a point $\mathbf{r}_s$ is purely normal in the Z -direction (the beam model neglects in-plane motion in the fluid coupling), so the plate velocity vector is

$$\mathbf{v}_s(X, \tau) = (0, \dot{W}(X, \tau)), \quad (47)$$

where

$$\dot{W}(X, \tau) = \frac{\partial W}{\partial \tau}(X, \tau). \quad (48)$$

The normal component of the plate velocity is

$$u_s(X, \tau) = \mathbf{v}_s(X, \tau) \cdot \mathbf{n}(X, \tau) = \dot{W}(X, \tau) n_Z(X, \tau). \quad (49)$$

The fluid velocity at the same point is

$$\mathbf{u}_f(X, \tau) = (U_\infty + \phi_X(X, W(X, \tau), t), \phi_Z(X, W(X, \tau), t)) \quad (50)$$

The kinematic boundary condition is that the normal component of fluid velocity equals the normal component of plate velocity at the plate:

$$\mathbf{u}_f(X, \tau) \cdot \mathbf{n}(X, \tau) = u_s(X, \tau). \quad (51)$$

Substituting the expressions above, we obtain

$$\begin{aligned} (U_\infty + \phi_X(X, W(X, \tau), t)) n_X(X, \tau) + \phi_Z(X, W(X, \tau), t) n_Z(X, \tau) \\ = \dot{W}(X, \tau) n_Z(X, \tau) \end{aligned} \quad (52)$$

This can be re-written as

$$u_n(X, \tau) + U_\infty n_X(X, \tau) = u_s(X, \tau). \quad (53)$$

with the fluid normal velocity from the perturbation potential as

$$u_n(X, \tau) = \phi_X(X, W(X, \tau), t) n_X(X, \tau) + \phi_Z(X, W(X, \tau), t) n_Z(X, \tau), \quad (54)$$

5.3 Boundary-Integral Representation / Discrete source method

To find the perturbation ϕ that generates $\phi_{,X}$ and $\phi_{,Z}$ that respect the BCs, we can use a continuous distribution of line sources along the plate. We introduce a source strength per unit length $\sigma(\zeta, t)$ defined along the plate parameter $\zeta \in [0, 1]$. The position of a generic source point on the plate is

$$\mathbf{r}_s(\zeta, \tau) = (\zeta, W(\zeta, \tau)). \quad (55)$$

For any point $\mathbf{r} = (X, Z)$ in the fluid domain, the distance vector and its magnitude are

$$\mathbf{R}(X, Z; \zeta, \tau) = \mathbf{r} - \mathbf{r}_s(\zeta, \tau) = (X - \zeta, Z - W(\zeta, \tau)), \quad (56)$$

$$R(X, Z; \zeta, \tau) = \sqrt{(X - \zeta)^2 + (Z - W(\zeta, \tau))^2}. \quad (57)$$

The perturbation potential generated by the sources is:

$$\phi(X, Z, t) = \frac{1}{2\pi} \int_0^1 \sigma(\zeta, t) \ln R(X, Z; \zeta, \tau) d\zeta. \quad (58)$$

We obtain the gradient of ϕ by differentiating under the integral sign. The gradient of $\ln R$ is

$$\nabla \ln R = \frac{1}{R^2} \mathbf{R}, \quad (59)$$

so we obtain

$$\nabla \phi(X, Z, t) = \frac{1}{2\pi} \int_0^1 \sigma(\zeta, t) \frac{\mathbf{R}(X, Z; \zeta, \tau)}{R^2(X, Z; \zeta, \tau)} d\zeta. \quad (60)$$

5.3.1 Evaluating $\nabla \phi$ on the plate

We now evaluate the normal component of $\nabla \phi$ on the plate, as required by Eq. 54. For a point on the plate at coordinate X , we set $Z = W(X, \tau)$ and define

$$\mathbf{r}_p(X, \tau) = (X, W(X, \tau)). \quad (61)$$

The distance vector from a source point at ζ to the plate point at X is

$$\mathbf{R}_p(X; \zeta, \tau) = \mathbf{r}_p(X, \tau) - \mathbf{r}_s(\zeta, \tau) = (X - \zeta, W(X, \tau) - W(\zeta, \tau)), \quad (62)$$

and its squared magnitude is

$$R_p^2(X; \zeta, \tau) = (X - \zeta)^2 + (W(X, \tau) - W(\zeta, \tau))^2. \quad (63)$$

We then write the gradient of ϕ on the plate as

$$\nabla \phi(X, W(X, \tau), t) = \frac{1}{2\pi} \int_0^1 \sigma(\zeta, t) \frac{\mathbf{R}_p(X; \zeta, \tau)}{R_p^2(X; \zeta, \tau)} d\zeta. \quad (64)$$

The normal component of this gradient is

$$u_n(X, \tau) = \nabla \phi(X, W(X, \tau), t) \cdot \mathbf{n}(X, \tau). \quad (65)$$

Substituting the expression for $\nabla \phi$, we obtain

$$u_n(X, \tau) = \frac{1}{2\pi} \int_0^1 \sigma(\zeta, t) \frac{\mathbf{R}_p(X; \zeta, \tau) \cdot \mathbf{n}(X, \tau)}{R_p^2(X; \zeta, \tau)} d\zeta. \quad (66)$$

We now substitute this expression into the kinematic condition of Eq. 53, obtaining the boundary-integral equation for the unknown source strength $\sigma(\zeta, t)$:

$$\frac{1}{2\pi} \int_0^1 \sigma(\zeta, t) \frac{\mathbf{R}_p(X; \zeta, \tau) \cdot \mathbf{n}(X, \tau)}{R_p^2(X; \zeta, \tau)} d\zeta + U_\infty n_X(X, \tau) = u_s(X, \tau). \quad (67)$$

5.3.2 Solution of the integral

The term in the integral

$$K_n(X, \zeta, \tau) = \frac{\mathbf{R}_p(X; \zeta, \tau) \cdot \mathbf{n}(X, \tau)}{R_p^2(X; \zeta, \tau)} \quad (68)$$

is singular as $\zeta \rightarrow X$. To isolate the singular part, we write

$$I_n(X, \zeta, \tau) = I_n^{\text{sing}}(X, \zeta, \tau) + I_n^{\text{reg}}(X, \zeta, \tau), \quad (69)$$

where I_n^{sing} is the singular part, evaluated when $X \rightarrow \zeta$, and I_n^{reg} is the integral term in the rest of the domain, which can be directly calculated numerically. The complete integral is therefore:

$$\frac{1}{2\pi} \int_0^1 \sigma(\zeta, t) K_n d\zeta = \frac{1}{2\pi} \int_0^{X-\Delta} \sigma(\zeta, t) K_n d\zeta + \frac{1}{2\pi} \int_{X+\Delta}^1 \sigma(\zeta, t) K_n d\zeta + I_n^{\text{sing}} \quad (70)$$

We consider the evaluation point X to be displaced infinitesimally from the boundary into the fluid domain by a distance ε , and subsequently take the limit $\varepsilon \rightarrow 0$. The integral of interest is the contribution of the singular term in the neighborhood of the singularity $\zeta \rightarrow X$:

$$I_n^{\text{sing}} = \lim_{\varepsilon \rightarrow 0} \frac{1}{2\pi} \int_{X-\Delta}^{X+\Delta} \sigma(\zeta) \frac{\mathbf{R} \cdot \mathbf{n}}{|\mathbf{R}|^2} d\zeta, \quad (71)$$

where Δ is a small interval around the singularity.

Letting the evaluation point P be offset from the plate surface $\mathbf{r}_p(X)$ along the unit normal $\mathbf{n}(X)$:

$$P = \mathbf{r}_p(X) + \varepsilon \mathbf{n}(X). \quad (72)$$

Using the explicit components $\mathbf{r}_p(X) = (X, W(X))$ and the normal vector defined before, $\mathbf{n} = \frac{1}{\sqrt{1+W_X^2}}(-W_X, 1)$, the coordinates of P are:

$$P = \left(X - \varepsilon \frac{W_X}{\sqrt{1+W_X^2}}, \quad W(X) + \varepsilon \frac{1}{\sqrt{1+W_X^2}} \right). \quad (73)$$

For the source position $S = \mathbf{r}_s(\zeta)$, we assume ζ is sufficiently close to X such that the curved plate can be locally linearized as a flat inclined plate. Letting $\delta = \zeta - X$, we approximate the curve as a straight line tangent to the plate:

$$S \approx (X + \delta, \quad W(X) + W_X \delta). \quad (74)$$

The distance vector $\mathbf{R}_p^*$ from the source S to the point P is $\mathbf{R}_p^* = P - S$. Its components are:

$$R_{pX}^* = -\delta - \varepsilon \frac{W_X}{\sqrt{1+W_X^2}}, \quad (75)$$

$$R_{pZ}^* = \frac{\varepsilon}{\sqrt{1+W_X^2}} - W_X \delta. \quad (76)$$

We now compute the numerator term $\mathbf{R}_p^* \cdot \mathbf{n}$ appearing in the kernel:

$$\begin{aligned} \mathbf{R}_p^* \cdot \mathbf{n} &= R_{pX}^* n_X + R_{pZ}^* n_Z = \left(-\delta - \frac{\varepsilon W_X}{\sqrt{1+W_X^2}} \right) \left(\frac{-W_X}{\sqrt{1+W_X^2}} \right) + \\ &+ \left(\frac{\varepsilon}{\sqrt{1+W_X^2}} - W_X \delta \right) \left(\frac{1}{\sqrt{1+W_X^2}} \right). \end{aligned} \quad (77)$$

Expanding the terms, the contributions involving δ cancel exactly:

$$\mathbf{R}_p^* \cdot \mathbf{n} = \frac{1}{(\sqrt{1+W_X^2})^2} (\varepsilon W_X^2 + \varepsilon) = \varepsilon. \quad (78)$$

and

$$|\mathbf{R}_p^*|^2 = \left(-\delta - \frac{\varepsilon W_X}{\sqrt{1+W_X^2}} \right)^2 + \left(\frac{\varepsilon}{\sqrt{1+W_X^2}} - W_X \delta \right)^2. \quad (79)$$

Upon expansion, the cross-terms $\pm 2\delta\varepsilon W_X / \sqrt{1+W_X^2}$ cancel, yielding:

$$|\mathbf{R}_p^*|^2 = \delta^2(1+W_X^2) + \frac{\varepsilon^2}{(\sqrt{1+W_X^2})^2}(1+W_X^2) = (\sqrt{1+W_X^2})^2 \delta^2 + \varepsilon^2. \quad (80)$$

Substituting these simplified expressions back into the integral, and assuming $\sigma(\zeta) \approx \sigma(X)$ is constant over the small interval $[-\Delta, \Delta]$:

$$I_{sing} = \frac{\sigma(X)}{2\pi} \int_{-\Delta}^{\Delta} \frac{\varepsilon}{\varepsilon^2 + (\sqrt{1+W_X^2})^2 \delta^2} d\delta. \quad (81)$$

This is a standard integral of the form $\int \frac{1}{a^2+u^2} du$ where we substitute $u = \sqrt{1+W_X^2}\delta$:

$$\begin{aligned} I_{sing} &= \frac{\sigma(X)}{2\pi\sqrt{1+W_X^2}} \left[\arctan\left(\frac{\sqrt{1+W_X^2}\delta}{\varepsilon}\right) \right]_{-\Delta}^{\Delta} = \\ &= \frac{\sigma(X)}{2\pi\sqrt{1+W_X^2}} \left[\arctan\left(\frac{\sqrt{1+W_X^2}\Delta}{\varepsilon}\right) - \arctan\left(\frac{-\sqrt{1+W_X^2}\Delta}{\varepsilon}\right) \right]. \end{aligned} \quad (82)$$

Taking the limit as $\varepsilon \rightarrow 0^+$, the arguments of the arc tangent function approach $\pm\infty$. Since $\arctan(\infty) = \pi/2$:

$$\lim_{\varepsilon \rightarrow 0} \left[\arctan\left(\frac{\sqrt{1+W_X^2}\Delta}{\varepsilon}\right) - \arctan\left(\frac{-\sqrt{1+W_X^2}\Delta}{\varepsilon}\right) \right] = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi. \quad (83)$$

The integral evaluates to:

$$I_{sing} = \frac{\sigma(X)}{2\sqrt{1+W_X^2}}. \quad (84)$$

5.4 Panel Discretization and Influence Matrices

We now discretize the plate into M panels. We introduce nodal points

$$0 = X_1 < X_2 < \dots < X_{M+1} = 1, \quad (85)$$

so that panel m spans the interval $[X_m, X_{m+1}]$. For each panel m , we define a control point at the panel center

$$X_m^c = \frac{X_m + X_{m+1}}{2}. \quad (86)$$

The corresponding plate displacement at the control point is

$$W_m(\tau) = W(X_m^c, \tau), \quad (87)$$

and the plate position vector at the control point is

$$\mathbf{r}_m(\tau) = (X_m^c, W_m(\tau)). \quad (88)$$

We approximate the source density $\sigma(\zeta, t)$ as piecewise constant on each panel:

$$\sigma(\zeta, t) \approx \sigma_m(t), \quad \zeta \in [X_m, X_{m+1}], \quad (89)$$

so we have one unknown $\sigma_m(t)$ per panel.

We write the integral equation at each control point $i = 1, 2, \dots, M$ by setting $X = X_i^c$ and approximating the integral as a sum of contributions from each panel. For each pair (i, m) , we define the normal influence coefficient

$$C_{im}^{(n)}(\tau) = \begin{cases} \frac{1}{2\pi} \int_{X_m}^{X_{m+1}} K_n(X_i^c, \zeta, \tau) d\zeta & \text{if } i \neq m \\ \frac{1}{2\sqrt{1+W_X^2}} & \text{if } i = m \end{cases} \quad (90)$$

We also define the normal plate velocity and the geometric projection of the uniform flow at the control point:

$$u_{s,i}(\tau) = u_s(X_i^c, \tau) = \dot{W}(X_i^c, \tau) n_{Z,i}(\tau), \quad (91)$$

$$n_{X,i}(\tau) = n_X(X_i^c, \tau). \quad (92)$$

With these definitions, the discrete form of the boundary-integral equation at each control point i is

$$\frac{1}{2\sqrt{1+W_{X_i}^2}} \sigma_i(t) + \sum_{i \neq m=1}^M C_{im}^{(n)}(\tau) \sigma_m(t) = u_{s,i}(\tau) - U_\infty n_{X,i}(\tau), \quad (93)$$

where $\sigma_i(t)$ denotes the source strength on panel i .

We write this system in matrix form. We define:

- The vector of source strengths

$$\boldsymbol{\sigma}(t) = (\sigma_1(t), \dots, \sigma_M(t))^T.$$

- The matrix

$$A_{im}(\tau) = \frac{1}{2\sqrt{1+W_{X_i}^2}} \delta_{im} + C_{im}^{(n)}(\tau),$$

where δ_{im} is the Kronecker delta.

- The right-hand-side vector

$$b_i(\tau) = u_{s,i}(\tau) - U_\infty n_{X,i}(\tau).$$

The discrete boundary-integral equation becomes

$$\sum_{m=1}^M A_{im}(\tau) \sigma_m(t) = b_i(\tau), \quad i = 1, \dots, M, \quad (94)$$

or in compact matrix form

$$A(\tau) \boldsymbol{\sigma}(t) = \mathbf{b}(\tau). \quad (95)$$

For a given plate shape $W(X, \tau)$ and plate velocity $\dot{W}(X, \tau)$, we compute $A(\tau)$ and $\mathbf{b}(\tau)$ and solve this linear system for $\boldsymbol{\sigma}(t)$.

5.5 Potential and its gradient at the plate

Once $\sigma(t)$ is known, we compute the perturbation velocity and potential at each control point on the plate by evaluating the boundary-integral representation.

We first write the perturbation potential at control point i as

$$\phi_i(t) = \frac{1}{2\pi} \sum_{m=1}^M \sigma_m(t) \int_{X_m}^{X_{m+1}} \ln R_i(\zeta, \tau) d\zeta = \sum_{m=1}^M I_{im}^{(\phi)}(\tau) \sigma_m(t). \quad (96)$$

where

$$R_i(\zeta, \tau) = \sqrt{(X_i^c - \zeta)^2 + (W_i(\tau) - W(\zeta, \tau))^2}. \quad (97)$$

and

$$I_{im}^{(\phi)}(\tau) = \frac{1}{2\pi} \int_{X_m}^{X_{m+1}} \ln R_i(\zeta, \tau) d\zeta, \quad (98)$$

Similarly, the perturbation velocity components at the control points come from:

$$\begin{aligned} \nabla \phi(X_i^c, W_i(\tau), t) &= (\phi_{X,i}(t), \phi_{Z,i}(t)) = \\ &= \frac{1}{2\pi} \int_0^1 \sigma(\zeta, t) \frac{\mathbf{R}_i(\zeta, \tau)}{R_i^2(\zeta, \tau)} d\zeta = \sum_{m=1}^M (I_{im}^{(u)}(\tau), I_{im}^{(v)}(\tau)) \sigma_m(t). \end{aligned} \quad (99)$$

where

$$\mathbf{R}_i(\zeta, \tau) = (X_i^c - \zeta, W_i(\tau) - W(\zeta, \tau)), \quad (100)$$

and

$$I_{im}^{(u)}(\tau) = \frac{1}{2\pi} \int_{X_m}^{X_{m+1}} \frac{X_i^c - \zeta}{R_i^2(\zeta, \tau)} d\zeta, \quad (101)$$

$$I_{im}^{(v)}(\tau) = \frac{1}{2\pi} \int_{X_m}^{X_{m+1}} \frac{W_i(\tau) - W(\zeta, \tau)}{R_i^2(\zeta, \tau)} d\zeta, \quad (102)$$

5.5.1 Evaluation of Self-Influence Coefficients

For the self-influence terms ($i = m$), the integrals contains a singularity at the control point. We define the panel half-width $\Delta = (X_{m+1} - X_m)/2$ and the local coordinate $\delta = \zeta - X_i^c$. Utilizing the flat panel approximation, as before, we have $R \approx |\delta| \sqrt{1 + W_{X,m}^2}$.

The potential influence coefficient becomes:

$$I_{mm}^{(\phi)} = \frac{1}{2\pi} \int_{-\Delta}^{\Delta} \ln \left(|\delta| \sqrt{1 + W_{X,m}^2} \right) d\delta \quad (103)$$

Since the integrand is an even function of δ , we integrate over the positive domain $[0, \Delta]$ and multiply by 2. Expanding the logarithm gives:

$$I_{mm}^{(\phi)} = \frac{1}{\pi} \left[\int_0^{\Delta} \ln \delta d\delta + \int_0^{\Delta} \ln \left(\sqrt{1 + W_{X,m}^2} \right) d\delta \right] \quad (104)$$

Using the integration identity $\int \ln x \, dx = x \ln x - x$, we evaluate the terms:

$$I_{mm}^{(\phi)} = \frac{1}{\pi} \left[(\Delta \ln \Delta - \Delta) + \Delta \ln \left(\sqrt{1 + W_{X,m}^2} \right) \right] \quad (105)$$

Factoring out Δ and combining the logarithmic terms yields the final analytical expression:

$$I_{mm}^{(\phi)} = \frac{\Delta}{\pi} \left[\ln \left(\Delta \sqrt{1 + W_{X,m}^2} \right) - 1 \right] \quad (106)$$

For the velocity influence coefficients $I_{mm}^{(u)}$ and $I_{mm}^{(w)}$, the integrands contain at the numerator the terms $(X_i^c - \zeta) = -\delta$ and $(W_i - W(\zeta)) \approx -W_{X,m}\delta$ in the numerator, respectively. Both integrands are odd functions of δ integrated over the symmetric interval $[-\Delta, \Delta]$. Therefore, their Cauchy principal values vanish:

$$I_{mm}^{(u)} = 0, \quad I_{mm}^{(w)} = 0 \quad (107)$$

Therefore the total fluid velocity at the plate control point i is then

$$u_i(t) = U_\infty + \phi_{X,i}(t), \quad (108)$$

$$w_{f,i}(t) = \phi_{Z,i}(t). \quad (109)$$

The matrix notation is:

$$\boldsymbol{\phi}(t) = I^{(\phi)}(\tau) \boldsymbol{\sigma}(t), \quad (110)$$

$$\boldsymbol{\phi}_X(t) = I^{(u)}(\tau) \boldsymbol{\sigma}(t), \quad \boldsymbol{\phi}_Z(t) = I^{(w)}(\tau) \boldsymbol{\sigma}(t), \quad (111)$$

where $\boldsymbol{\phi}(t)$, $\boldsymbol{\phi}_X(t)$, and $\boldsymbol{\phi}_Z(t)$ are the vectors collecting $\phi_i(t)$, $\phi_{X,i}(t)$, and $\phi_{Z,i}(t)$ for $i = 1, \dots, M$.

5.6 Unsteady Bernoulli Equation and Pressure on the Plate

We now relate the perturbation potential and velocities at the plate to the pressure on the plate using the unsteady Bernoulli equation. We recall the structural scaling already in use:

$$x = LX, \quad z = LZ, \quad t = t_s \tau, \quad t_s = \sqrt{\frac{mL^4}{EI}},$$

$$W(X, \tau) = \frac{w(x, t)}{L}.$$

We use the same scales to non-dimensionalize the fluid potential and velocities. Let $\tilde{\Phi}(x, z, t)$ and $\tilde{p}(x, t)$ denote the dimensional velocity potential and pressure. We define the non-dimensional potential $\Phi(X, Z, \tau)$ by

$$\tilde{\Phi}(x, z, t) = \Phi_s \Phi(X, Z, \tau), \quad \Phi_s = \frac{L^2}{t_s},$$

so that the characteristic velocity scale is

$$U_s = \frac{L}{t_s}.$$

We introduce the structurally non-dimensional pressure

$$p(X, \tau) = \frac{L^3}{EI} \tilde{p}(x, t),$$

and define the dimensionless mass ratio between structure and fluid

$$\beta = \frac{m}{\rho_f L},$$

where m is the structural mass per unit length and ρ_f is the fluid density.

The dimensional unsteady Bernoulli equation in an inertial frame is

$$\frac{\partial \tilde{\Phi}}{\partial t} + \frac{1}{2}(\tilde{u}^2 + \tilde{w}_f^2) + \frac{\tilde{p}}{\rho_f} = B_{\text{dim}}(t),$$

with $B_{\text{dim}}(t)$ uniform in space,. In the non-dimensional form:

$$\frac{\partial \tilde{\Phi}}{\partial t} = \frac{EI}{mL^2} \frac{\partial \Phi}{\partial \tau},$$

$$\frac{1}{2}(\tilde{u}^2 + \tilde{w}_f^2) = \frac{1}{2} \frac{EI}{mL^2} (U^2 + W_f^2),$$

Substituting into Bernoulli and dividing by $EI/(mL^2)$ gives

$$\frac{\partial \Phi}{\partial \tau} + \frac{1}{2}(U^2 + W_f^2) + \frac{mL^2}{EI} \frac{\tilde{p}}{\rho} = B(\tau),$$

where $B(\tau) = B_{\text{dim}}(t) mL^2/EI$.

Using $\tilde{p} = (EI/L^3)p$, the pressure term becomes

$$\frac{mL^2}{EI} \frac{\tilde{p}}{\rho} = \frac{mL^2}{EI} \frac{1}{\rho} \frac{EI}{L^3} p = \frac{m}{\rho L} p = \beta p,$$

so the non-dimensional Bernoulli equation in structural variables is

$$\frac{\partial \Phi}{\partial \tau} + \frac{1}{2}(U^2 + W_f^2) + \beta p = B(\tau). \quad (112)$$

For the uniform base flow

$$\Phi_0(X) = U_\infty X, \quad U = U_\infty, \quad W_f = 0, \quad p = p_\infty(\tau),$$

Bernoulli gives

$$\frac{1}{2}U_\infty^2 + \beta p_\infty(\tau) = B(\tau). \quad (113)$$

We subtract this base-flow relation from the full Bernoulli equation. Using $\Phi = \Phi_0 + \phi$, with Φ_0 steady, we have $\partial \Phi / \partial \tau = \partial \phi / \partial \tau$, and

$$U^2 + W_f^2 = (U_\infty + \phi_X)^2 + \phi_Z^2.$$

The difference gives

$$\frac{\partial \phi}{\partial \tau} + \frac{1}{2} [(U_\infty + \phi_X)^2 + \phi_Z^2 - U_\infty^2] + \beta(p - p_\infty) = 0. \quad (114)$$

We define the structurally non-dimensional perturbation pressure

$$\delta p(X, \tau) = p(X, \tau) - p_\infty(\tau),$$

so that

$$\delta p(X, \tau) = -\frac{1}{\beta} \left[\frac{\partial \phi}{\partial \tau}(X, W(X, \tau), \tau) + \frac{1}{2} ((U_\infty + \phi_X)^2 + \phi_Z^2 - U_\infty^2) \right]. \quad (115)$$

We will take δp as the pressure loading on the plate, i.e. we measure pressure relative to the uniform upstream value p_∞ .

At the discrete level, evaluating at panel control point i :

$$\delta p_i(\tau) = -\frac{1}{\beta} \left[\dot{\phi}_i(\tau) + \frac{1}{2} ((U_\infty + \phi_{X,i}(\tau))^2 + \phi_{Z,i}^2(\tau) - U_\infty^2) \right], \quad (116)$$

5.7 Projection of the Fluid Pressure onto the Plate Modes

We now project the non-dimensional fluid pressure onto the Galerkin modes of the bistable plate, in order to obtain modal forcing terms consistent with the structural equations. We define the non-dimensional generalized force

$$f_k(\tau) = \frac{L^2}{EI} F_k(t(\tau)),$$

so that

$$f_k(\tau) = - \int_0^1 p(X, \tau) \phi_k(X) dX. \quad (117)$$

With this definition, the non-dimensional equation of motion for mode k becomes

$$\ddot{q}_k(\tau) + \lambda_k^4 q_k(\tau) + \alpha \left(-\bar{\Delta} + \frac{1}{2} \sum_{i,j} q_i(\tau) q_j(\tau) B_{ij} \right) \sum_i q_i(\tau) B_{ik} = f_k(\tau), \quad (118)$$

If we assume that $\bar{p}(X, \tau)$ is approximately constant on panel i and equal to $\bar{p}_i(\tau)$, then for panel i with non-dimensional length

$$\Delta X_i = X_{i+1} - X_i,$$

we have

$$f_k(\tau) \approx - \sum_{i=1}^M p_i(\tau) \phi_k(X_i^c) \Delta X_i. \quad (119)$$

5.8 Decomposition of Fluid Forces and Added Mass

We now separate the fluid forces acting on the plate into inertial terms (added mass), damping terms, and convective forcing terms. This allows us to couple the fluid dynamics directly into the Galerkin generalized coordinates of the structure.

Starting with the non-dimensional perturbation pressure $\delta p_i(\tau)$ at the i -th control point, as given by the discrete unsteady Bernoulli equation:

$$\delta p_i(\tau) = -\frac{1}{\beta} \left[\dot{\phi}_i(\tau) + \frac{1}{2} \left((U_\infty + \phi_{X,i}(\tau))^2 + \phi_{Z,i}^2(\tau) - U_\infty^2 \right) \right]. \quad (120)$$

Our primary objective is to expand the term $\dot{\phi}_i(\tau)$ (the time derivative of the potential) to isolate terms proportional to the structural acceleration $\ddot{q}$ and velocity $\dot{q}$.

The perturbation potential vector $\phi(\tau)$ is related to the source strength vector $\sigma(\tau)$ by the potential influence matrix $I^{(\phi)}(\tau)$:

$$\phi(\tau) = I^{(\phi)}(\tau)\sigma(\tau). \quad (121)$$

The source strength $\sigma(\tau)$ is the solution to the linear boundary integral system $A(\tau)\sigma(\tau) = \mathbf{b}(\tau)$. Therefore, $\sigma(\tau) = A^{-1}(\tau)\mathbf{b}(\tau)$. Substituting this into the potential equation, we define the matrix $\mathbb{T}(\tau)$ as:

$$\mathbb{T}(\tau) = I^{(\phi)}(\tau)A^{-1}(\tau). \quad (122)$$

Thus, the potential vector is given by:

$$\phi(\tau) = \mathbb{T}(\tau)\mathbf{b}(\tau). \quad (123)$$

For the i -th element, $b_i(\tau) = u_{s,i}(\tau) - U_\infty n_{X,i}(\tau)$. Using the Galerkin expansion $W(X, \tau) = \sum_{k=1}^N q_k(\tau)\phi_k(X)$, the structural velocity is $\dot{W} = \sum \dot{q}_k\phi_k$. We decompose $\mathbf{b}$ into a velocity-dependent part and a convective part:

$$\mathbf{b}(\tau) = \sum_{k=1}^N \dot{q}_k(\tau)\mathbf{V}_k(\tau) + \mathbf{b}_{conv}(\tau), \quad (124)$$

where the components of the vector $\mathbf{V}_k$ and the convective vector $\mathbf{b}_{conv}$ are:

$$(\mathbf{V}_k)_i = \phi_k(X_i^c)n_{Z,i}(\tau), \quad (\mathbf{b}_{conv})_i = -U_\infty n_{X,i}(\tau). \quad (125)$$

5.8.1 Expansion of the Time Derivative $\dot{\phi}$

Substituting the decomposed boundary vector into the potential equation yields:

$$\phi(\tau) = \sum_{k=1}^N \dot{q}_k(\tau)\mathbb{T}\mathbf{V}_k + \mathbb{T}\mathbf{b}_{conv}. \quad (126)$$

Differentiating with respect to time τ , and noting that matrices depend on time via the generalized coordinates $q(\tau)$:

$$\dot{\phi} = \underbrace{\sum_{k=1}^N \ddot{q}_k (\mathbb{T} \mathbf{V}_k)}_{\text{Acceleration Terms}} + \underbrace{\sum_{k=1}^N \dot{q}_k \frac{d}{d\tau} (\mathbb{T} \mathbf{V}_k) + \frac{d}{d\tau} (\mathbb{T} \mathbf{b}_{conv})}_{\text{Velocity Terms}}. \quad (127)$$

We define $\psi_k(\tau) = \mathbb{T}(\tau) \mathbf{V}_k(\tau)$, and the remaining terms as $\dot{\phi}_{damp}$. The derivative becomes:

$$\dot{\phi} = \sum_{k=1}^N \ddot{q}_k \psi_k + \dot{\phi}_{damp}. \quad (128)$$

5.8.2 Computation of Damping Terms from $\dot{\phi}_{damp}$

To compute the term $\dot{\phi}_{damp}$ explicitly for the Galerkin decomposed EOM, we want to write it as proportional to $\dot{q}_j$.

The damping potential rate is defined as:

$$\dot{\phi}_{damp} = \sum_{k=1}^N \dot{q}_k \dot{\psi}_k + \frac{d}{d\tau} (\mathbb{T} \mathbf{b}_{conv}). \quad (129)$$

Using the identity $\frac{d(\cdot)}{d\tau} = \sum_{j=1}^N \frac{\partial(\cdot)}{\partial q_j} \dot{q}_j$, we expand the components:

$$\dot{\psi}_k = \sum_{j=1}^N \left(\frac{\partial \mathbb{T}}{\partial q_j} \mathbf{V}_k + \mathbb{T} \frac{\partial \mathbf{V}_k}{\partial q_j} \right) \dot{q}_j, \quad (130)$$

$$\frac{d}{d\tau} (\mathbb{T} \mathbf{b}_{conv}) = \sum_{j=1}^N \left(\frac{\partial \mathbb{T}}{\partial q_j} \mathbf{b}_{conv} + \mathbb{T} \frac{\partial \mathbf{b}_{conv}}{\partial q_j} \right) \dot{q}_j. \quad (131)$$

The partial derivatives of the normal vector components at control point i with respect to q_j are, using their definition:

$$\frac{\partial n_{Z,i}}{\partial q_j} = n_{X,i} n_{Z,i}^2 \phi_j'(X_i^c), \quad \frac{\partial n_{X,i}}{\partial q_j} = -n_{Z,i}^3 \phi_j'(X_i^c). \quad (132)$$

Consequently:

$$\left(\frac{\partial \mathbf{V}_k}{\partial q_j} \right)_i = \phi_k(X_i^c) [n_{X,i} n_{Z,i}^2 \phi_j'(X_i^c)], \quad (133)$$

$$\left(\frac{\partial \mathbf{b}_{conv}}{\partial q_j} \right)_i = U_\infty [n_{Z,i}^3 \phi_j'(X_i^c)]. \quad (134)$$

The derivative of the transfer matrix $\mathbb{T} = I^{(\phi)} A^{-1}$ is given by:

$$\frac{\partial \mathbb{T}}{\partial q_j} = \left(\frac{\partial I^{(\phi)}}{\partial q_j} - \mathbb{T} \frac{\partial A}{\partial q_j} \right) A^{-1}. \quad (135)$$

$\frac{\partial I^{(\phi)}}{\partial q_j}$. For off-diagonal terms ($i \neq m$), differentiating the integral of $\ln R$ yields:

$$\frac{\partial I_{im}^{(\phi)}}{\partial q_j} = \frac{1}{2\pi} \int_{X_m}^{X_{m+1}} \frac{(W_i - W(\zeta))(\phi_j(X_i^c) - \phi_j(\zeta))}{(X_i^c - \zeta)^2 + (W_i - W(\zeta))^2} d\zeta. \quad (136)$$

For the self-influence term ($i = m$), using the analytical expression:

$$\frac{\partial I_{mm}^{(\phi)}}{\partial q_j} = \frac{\Delta}{\pi} \frac{W_{X,m} \phi_j'(X_m^c)}{1 + W_{X,m}^2}. \quad (137)$$

$\frac{\partial A}{\partial q_j}$. The matrix is $A_{im} = D_{ii} \delta_{im} + C_{im}^{(n)}$. The derivative of the diagonal term is:

$$\frac{\partial D_{ii}}{\partial q_j} = -\frac{W_{X,i} \phi_j'(X_i^c)}{2(1 + W_{X,i}^2)^{3/2}}. \quad (138)$$

The derivative of $C_{im}^{(n)}$ is:

$$\frac{\partial C_{im}^{(n)}}{\partial q_j} = \frac{1}{2\pi} \int_{X_m}^{X_{m+1}} \frac{\partial K_n}{\partial q_j} d\zeta. \quad (139)$$

Where, for $i \neq m$:

$$\frac{\partial K_n}{\partial q_j} = \frac{1}{|\mathbf{R}_p|^2} \left(\frac{\partial \mathbf{R}_p}{\partial q_j} \cdot \mathbf{n}_i + \mathbf{R}_p \cdot \frac{\partial \mathbf{n}_i}{\partial q_j} \right) - \frac{2(\mathbf{R}_p \cdot \mathbf{n}_i)}{|\mathbf{R}_p|^4} \left(\mathbf{R}_p \cdot \frac{\partial \mathbf{R}_p}{\partial q_j} \right), \quad (140)$$

where $\frac{\partial \mathbf{R}_p}{\partial q_j} = (0, \phi_j(X_i^c) - \phi_j(\zeta))$.

5.8.3 Final Assembly

Remembering from before:

$$\dot{\phi} = \sum_{k=1}^N \ddot{q}_k \psi_k + \dot{\phi}_{damp}. \quad (141)$$

Projecting the pressure onto the Galerkin modes, the generalized fluid force is $f_k = -\sum \delta p_i \phi_k(X_i^c) \Delta X_i$.

Separating the acceleration terms in the Bernoulli equation, we can define an additional mass term M_{kj}^{fluid} as:

$$M_{kj}^{fluid}(\tau) = -\frac{1}{\beta} \sum_{i=1}^M \psi_{j,i} \phi_k(X_i^c) \Delta X_i. \quad (142)$$

The total damping potential rate is computed by summing these contributions:

$$\dot{\phi}_{damp} = \sum_{j=1}^N \dot{q}_j \left[\frac{\partial \mathbb{T}}{\partial q_j} \mathbf{b}_{conv} + \mathbb{T} \frac{\partial \mathbf{b}_{conv}}{\partial q_j} + \sum_{k=1}^N \dot{q}_k \left(\frac{\partial \mathbb{T}}{\partial q_j} \mathbf{V}_k + \mathbb{T} \frac{\partial \mathbf{V}_k}{\partial q_j} \right) \right]. \quad (143)$$

so we can define a linear and a quadratic damping-like term $C_{kj}^{1,fluid}$ and $C_{kj}^{2,fluid}$ as:

$$C_{kj}^{1,fluid}(\tau) = -\frac{1}{\beta} \sum_{i=1}^M \left[\frac{\partial \mathbb{T}}{\partial q_j} \mathbf{b}_{conv} + \mathbb{T} \frac{\partial \mathbf{b}_{conv}}{\partial q_j} \right]_i \phi_k(X_i^c) \Delta X_i. \quad (144)$$

$$C_{kjl}^{2,fluid}(\tau) = -\frac{1}{\beta} \sum_{i=1}^M \left[\left(\frac{\partial \mathbb{T}}{\partial q_j} \mathbf{v}_l + \mathbb{T} \frac{\partial \mathbf{v}_l}{\partial q_j} \right) \right]_i \phi_k(X_i^c) \Delta X_i. \quad (145)$$

The remaining force proportional to the plate shape is:

$$f_k^{conv}(\tau) = \frac{1}{\beta} \sum_{i=1}^M \left[\frac{1}{2} ((U_\infty + \phi_{X,i}(\tau))^2 + \phi_{Z,i}^2(\tau) - U_\infty^2) \right] \phi_k(X_i^c) \Delta X_i, \quad (146)$$

The final coupled equation of motion, moving the added mass to the left-hand side, is:

$$\left(\delta_{kj} + M_{kj}^{fluid}(\tau) \right) \ddot{q}_j + C_{kj}^{fluid}(\tau) \dot{q}_j + C_{kjl}^{2,fluid}(\tau) \dot{q}_j \dot{q}_l + \lambda_k^4 q_k + \alpha S \sum_i q_i B_{ik} = f_k^{conv}(\tau). \quad (147)$$

6 Fluid-Vortex-Structure Interaction

To capture the physical interaction where the bistable plate creates a blockage that deflects the incoming wake (and conversely, where the wake forces the plate to snap), we discard the imposed "frozen" vortex street model. Instead, we implement a Discrete Vortex Method (DVM) with a free wake. In this framework, vortices are Lagrangian particles released upstream that are advected by the local velocity field, which is the superposition of the uniform flow, the potential flow generated by the plate's deformation, and the mutual induction of the vortices themselves.

6.1 Discrete Vortex Representation

We model the wake as a collection of $N_v(\tau)$ discrete point vortices. The k -th vortex is defined by its position vector $\mathbf{x}_k(\tau) = (\xi_k(\tau), \zeta_k(\tau))$ and its constant circulation Γ_k . To satisfy the impermeability condition of the rigid wall at $Z = 0$, we employ the method of images. For every physical vortex at (ξ_k, ζ_k) with strength Γ_k , there exists an image vortex at $(\xi_k, -\zeta_k)$ with strength $-\Gamma_k$.

The velocity field at any point $\mathbf{x} = (X, Z)$ in the fluid is decomposed into three components:

$$\mathbf{u}(\mathbf{x}, \tau) = \mathbf{u}_\infty + \mathbf{u}_{plate}(\mathbf{x}, \tau) + \mathbf{u}_{vort}(\mathbf{x}, \tau) \quad (148)$$

where:

- $\mathbf{u}_\infty = (U_\infty, 0)$ is the uniform freestream velocity.
- $\mathbf{u}_{plate}$ is the perturbation velocity induced by the distributed sources σ on the plate (representing the structural deformation and motion).
- $\mathbf{u}_{vort}$ is the velocity induced by the discrete vortices and their images.

Using the Biot-Savart law in 2D, the velocity induced by the vortex system at a point $\mathbf{x}$ is:

$$\mathbf{u}_{vort}(\mathbf{x}) = \sum_{k=1}^{N_v} \frac{\Gamma_k}{2\pi} \left[\frac{\hat{\mathbf{y}} \times (\mathbf{x} - \mathbf{x}_k)}{|\mathbf{x} - \mathbf{x}_k|^2} - \frac{\hat{\mathbf{y}} \times (\mathbf{x} - \mathbf{x}_k^{img})}{|\mathbf{x} - \mathbf{x}_k^{img}|^2} \right] \quad (149)$$

where $\mathbf{x}_k^{img} = (\xi_k, -\zeta_k)$ and $\hat{\mathbf{y}}$ is the unit vector out of the plane.

6.2 Two-Way Coupling Mechanism

The interaction is coupled through the kinematic boundary condition on the plate surface and the transport equations for the vortices.

6.2.1 Vortex Influence on the Plate

The presence of vortices modifies the inflow conditions seen by the plate. The kinematic boundary condition requires that the total normal fluid velocity equals the structural normal velocity u_s .

$$(\mathbf{u}_\infty + \mathbf{u}_{plate} + \mathbf{u}_{vort}) \cdot \mathbf{n} = u_s \quad (150)$$

Rearranging terms, the source distribution $\sigma(\tau)$ on the plate must satisfy:

$$\mathbf{u}_{plate}(\sigma) \cdot \mathbf{n} = u_s - U_\infty n_X - (\mathbf{u}_{vort} \cdot \mathbf{n}) \quad (151)$$

In the discretized panel method, this modifies the right-hand side vector $b(\tau)$ of the linear system $A\sigma = b$. The i -th component becomes:

$$b_i(\tau) = u_{s,i}(\tau) - U_\infty n_{X,i}(\tau) - [u_{v,i} n_{X,i}(\tau) + w_{v,i} n_{Z,i}(\tau)] \quad (152)$$

where $(u_{v,i}, w_{v,i})$ is the vortex-induced velocity vector evaluated at the i -th control point.

The motion of the vortices is governed by the Helmholtz theorems, which state that vortex lines move with the local fluid velocity. The Lagrangian equations of motion for the k -th vortex are:

$$\frac{d\xi_k}{d\tau} = u_{total}(\mathbf{x}_k, \tau), \quad \frac{d\zeta_k}{d\tau} = w_{total}(\mathbf{x}_k, \tau) \quad (153)$$

The velocity induced by the plate at the location of the k -th vortex is computed by summing the contributions of all M panels:

$$\mathbf{u}_{plate}(\mathbf{x}_k) = \sum_{m=1}^M \frac{\sigma_m \Delta X_m}{2\pi} \frac{\mathbf{x}_k - \mathbf{x}_{c,m}}{|\mathbf{x}_k - \mathbf{x}_{c,m}|^2} \quad (154)$$

6.3 Vortex Shedding and Dynamics

Instead of a pre-existing street, vortices are shed dynamically from an upstream source location (ξ_{src}, ζ_{src}) to simulate the wake of a bluff body (cylinder) of diameter D .

1. **Shedding Frequency:** New vortices are introduced at time intervals $\Delta\tau_s = (St \cdot U_\infty / D)^{-1}$, where $St \approx 0.21$ is the Strouhal number.
2. **Circulation:** The circulation strength is determined by the rate of vorticity flux, approximated as $\Gamma_{shed} \approx 0.5 U_\infty^2 \Delta\tau_s$. The sign alternates $\pm \Gamma_{shed}$ for each release. The circulation alternates direction between consecutive vortices. The first vortex ($k = 1$) has negative circulation (clockwise):

$$\Gamma_{shed,k} = (-1)^k \Gamma_{shed} \quad (155)$$

3. **Vertical Position (ζ_k):** The vortices alternate between the upper and lower rows of the street. The upper row corresponds to negative circulation (k odd), and the lower row to positive circulation (k even):

$$\zeta_k = \begin{cases} Z_c + D/2 & \text{for } k \text{ odd} \\ Z_c - D/2 & \text{for } k \text{ even} \end{cases} \quad (156)$$

4. **Regularization:** To prevent numerical singularities during close encounters (vortex-vortex or vortex-plate interactions), the singular kernel $1/R$ in Eq. 149 is replaced by a desingularized core model (e.g., Rankine or Vatistas core) with radius δ :

$$\frac{1}{R} \rightarrow \frac{R}{R^2 + \delta^2} \quad (157)$$

6.4 Modified Unsteady Pressure

The pressure loading on the plate is derived from the unsteady Bernoulli equation. The term $\partial\phi/\partial\tau$ must account for the time-dependent potential of the moving vortices. The potential of the vortex system is:

$$\phi_{vort}(\mathbf{x}, \tau) = \sum_{k=1}^{N_v} \frac{\Gamma_k}{2\pi} \left[\arctan \frac{Z - \zeta_k(\tau)}{X - \xi_k(\tau)} - \arctan \frac{Z + \zeta_k(\tau)}{X - \xi_k(\tau)} \right] \quad (158)$$

The time derivative is computed via the chain rule using the vortex velocities ($\dot{\xi}_k, \dot{\zeta}_k$):

$$\frac{\partial\phi_{vort}}{\partial\tau} = \sum_{k=1}^{N_v} \left(\frac{\partial\phi_{vort}}{\partial\xi_k} \dot{\xi}_k + \frac{\partial\phi_{vort}}{\partial\zeta_k} \dot{\zeta}_k \right) = -\mathbf{u}_{vort}(\mathbf{x}) \cdot \mathbf{u}_k \quad (159)$$

This term is added to the potential rate $\dot{\phi}_{plate}$ (from the variation of σ) in the pressure equation:

$$\delta p_i(\tau) = -\frac{1}{\beta} \left[\dot{\phi}_{plate,i} + \left(\frac{\partial\phi_{vort}}{\partial\tau} \right)_i + \frac{1}{2} (|\mathbf{u}_{total,i}|^2 - U_\infty^2) \right] \quad (160)$$

This formulation ensures that the pressure transients caused by a vortex passing close to the plate surface are accurately captured, providing the forcing mechanism necessary to trigger snap-through buckling.

7 Beam-fluid interaction results

7.1 Stability analysis in uniform flow

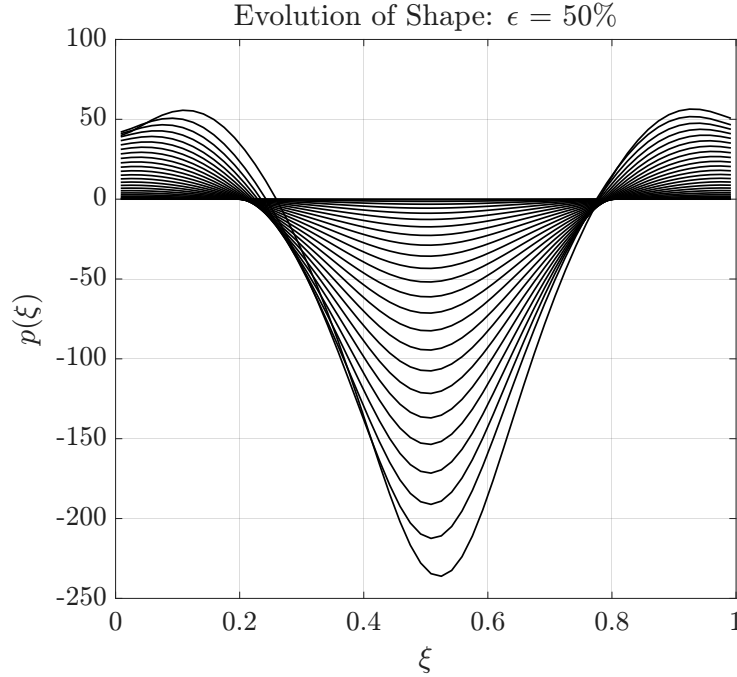


Figure 9: Pressure profile over plate. The pressure has a strong Mode 3-like component. At the edges the plate is pushed down, while a lifting force is present at the center.

To systematically determine the dynamics of the plate without running time-domain simulations for every case, we perform a linear stability analysis of the static equilibrium. We define the static equilibrium by setting the time-dependent terms $(\ddot{q}, \dot{q})$ in the coupled equation of motion to zero. The equilibrium state q_{eq} satisfies the nonlinear algebraic system $R_k(q) = 0$ for each mode k :

$$R_k(q) = F_k^{struct}(q) - f_k^{conv}(q) = 0 \quad (161)$$

where F_k^{struct} represents the internal structural restoring forces and f_k^{conv} represents the external steady fluid forcing. The stability of this equilibrium is determined by the eigenvalues of the Tangent Stiffness Matrix K_T , which is the Jacobian of the residual vector with respect to the generalized coordinates q_j :

$$(K_T)_{kj} = \frac{\partial R_k}{\partial q_j} = \frac{\partial F_k^{struct}}{\partial q_j} - \frac{\partial f_k^{conv}}{\partial q_j} \quad (162)$$

A positive definite K_T implies a stable equilibrium. If the lowest eigenvalue crosses zero, the system undergoes a divergence instability (snap-through). We derive the structural and fluid contributions separately.

Structural Tangent Stiffness The internal force for mode k is given by the static part of the Galerkin projection:

$$F_k^{struct} = \lambda_k^4 q_k + \alpha S(q) \sum_{i=1}^N q_i B_{ik} \quad (163)$$

where the non-dimensional axial tension parameter $S(q)$ is:

$$S(q) = -\bar{\Delta} + \frac{1}{2} \sum_{m=1}^N \sum_{n=1}^N q_m q_n B_{mn} \quad (164)$$

Differentiating F_k^{struct} with respect to a perturbation ∂q_j , and utilizing the symmetry of the integral coefficients ($B_{ij} = B_{ji}$), yields two components: the variation of the displacement field and the variation of the membrane tension S . The derivative of the tension is:

$$\frac{\partial S}{\partial q_j} = \frac{1}{2} \sum_{m,n} (\delta_{mj} q_n B_{mn} + q_m \delta_{nj} B_{mn}) = \sum_{m=1}^N q_m B_{jm} \quad (165)$$

Substituting this back into the total derivative, we obtain the structural tangent stiffness matrix:

$$(K_{struct})_{kj} = \lambda_k^4 \delta_{kj} + \alpha \left[\left(\sum_i q_i B_{ik} \right) \left(\sum_m q_m B_{jm} \right) + S B_{kj} \right] \quad (166)$$

The term $\alpha S B_{kj}$ represents the stress-stiffening effect, which decreases as the plate is pushed downwards (lowering S), eventually leading to loss of structural stability.

Aerodynamic Stiffness The fluid contribution arises from the variation of the convective pressure forces as the plate deforms. The generalized convective force is:

$$f_k^{conv} = \frac{1}{\beta} \sum_{p=1}^M \frac{1}{2} [(U_\infty + \phi_{X,p})^2 + \phi_{Z,p}^2 - U_\infty^2] \phi_k(X_p^c) \Delta X_p \quad (167)$$

Differentiating with respect to q_j , we apply the chain rule to the velocity components $\phi_{X,p}$ and $\phi_{Z,p}$:

$$(K_{fluid})_{kj} = \frac{\partial f_k^{conv}}{\partial q_j} = \frac{1}{\beta} \sum_{p=1}^M \left[(U_\infty + \phi_{X,p}) \frac{\partial \phi_{X,p}}{\partial q_j} + \phi_{Z,p} \frac{\partial \phi_{Z,p}}{\partial q_j} \right] \phi_k(X_p^c) \Delta X_p \quad (168)$$

To evaluate the velocity sensitivities $\frac{\partial \phi_{X,p}}{\partial q_j}$ and $\frac{\partial \phi_{Z,p}}{\partial q_j}$, we recall that the velocities are related to the source strengths σ via influence matrices. For the horizontal component (and similarly for vertical):

$$\frac{\partial \phi_{X,p}}{\partial q_j} = \sum_{m=1}^M \left(\frac{\partial I_{pm}^{(u)}}{\partial q_j} \sigma_m + I_{pm}^{(u)} \frac{\partial \sigma_m}{\partial q_j} \right) \quad (169)$$

The sensitivity of the source strengths, $\frac{\partial \sigma}{\partial q_j}$, is found by differentiating the boundary integral system $A\sigma = b$:

$$\frac{\partial \sigma}{\partial q_j} = A^{-1} \left(\frac{\partial b_{conv}}{\partial q_j} - \frac{\partial A}{\partial q_j} \sigma \right) \quad (170)$$

The component derivatives required for this assembly have been derived in Section 6.8.2. We report the final expressions here for completeness:

$$\left(\frac{\partial b_{conv}}{\partial q_j} \right)_i = U_\infty n_{Z,i}^3 \phi_j'(X_i^c) \quad (171)$$

The diagonal and off-diagonal terms are differentiated separately:

$$\frac{\partial A_{ii}}{\partial q_j} = -\frac{W_{X,i} \phi_j'(X_i^c)}{2(1 + W_{X,i}^2)^{3/2}} \quad (172)$$

$$\frac{\partial A_{im}}{\partial q_j} = \frac{1}{2\pi} \int_{X_m}^{X_{m+1}} \frac{\partial K_n}{\partial q_j} d\zeta \quad (i \neq m) \quad (173)$$

$$\frac{\partial I_{im}^{(\phi)}}{\partial q_j} = \frac{1}{2\pi} \int_{X_m}^{X_{m+1}} \frac{(W_i - W(\zeta))(\phi_j(X_i^c) - \phi_j(\zeta))}{(X_i^c - \zeta)^2 + (W_i - W(\zeta))^2} d\zeta \quad (174)$$

$$\frac{\partial I_{mm}^{(\phi)}}{\partial q_j} = \frac{\Delta}{\pi} \frac{W_{X,m} \phi_j'(X_m^c)}{1 + W_{X,m}^2} \quad (175)$$

Total Tangent Stiffness and Stability Criterion The final tangent stiffness matrix is assembled as:

$$K_T(q_{eq}, U_\infty) = K_{struct}(q_{eq}) - K_{fluid}(q_{eq}, U_\infty) \quad (176)$$

Stability is assessed by computing the eigenvalues μ of K_T . The condition $\mu_{min} > 0$ indicates a stable configuration. The threshold velocity for snap-down is identified when $\det(K_T) = 0$.

7.2 Stability validation

To verify the accuracy of the coupled fluid-structure interaction framework, the linear aeroelastic stability of the model is validated against the classical theoretical threshold for static divergence of a clamped-clamped panel, as established by Kornecky, Dowell, and O'Brien [?].

In their classical analysis, the authors demonstrated that a two-dimensional panel fixed at both the leading and trailing edges and exposed to an incompressible potential flow loses stability via static divergence rather than flutter. To reproduce this result using the present generalized formulation, the bistable mechanics are temporarily bypassed to enforce a perfectly flat equilibrium state. Specifically, the mechanical pre-compression is removed ($\bar{\Delta} = 0$) and the initial geometric imperfection is set to zero ($\epsilon = 0$).

The stability of the flat equilibrium state $q_{eq} = \mathbf{0}$ is assessed by tracking the eigenvalues of the total tangent stiffness matrix K_T as the flow velocity increases. The fluid's steady aerodynamic pressure acts as a velocity-dependent negative stiffness that counteracts the structural bending stiffness. Static divergence occurs at the critical velocity where the aerodynamic stiffness overwhelms the structural stiffness, causing the fundamental natural frequency of the system to drop to zero.

Dowell defines the non-dimensional dynamic pressure parameter λ as:

$$\lambda^2 = \frac{\rho_f U_{phys}^2 L^3}{EI} \quad (177)$$

where ρ_f is the fluid density, U_{phys} is the physical flow velocity, L is the plate length, and EI is the flexural rigidity (equivalent to Dowell's D). For a clamped-clamped panel, the exact theoretical threshold for static divergence is derived to be $\lambda_{cr}^2 \approx 181$ [?].

In the present model, the non-dimensional flow velocity $\bar{U}_\infty$ is scaled by the characteristic structural velocity $U_s = \sqrt{EI/(mL^2)}$, where m is the structural mass per unit length. The conversion between the two velocities is obtained by substituting U_{phys} from Dowell's definition:

$$\bar{U}_\infty^2 = \frac{U_{phys}^2}{U_s^2} = \left(\lambda^2 \frac{EI}{\rho_f L^3} \right) \left(\frac{mL^2}{EI} \right) = \lambda^2 \left(\frac{m}{\rho_f L} \right)$$

Recognizing that the mass ratio in our formulation is defined as $\beta = m/(\rho_f L)$, the conversion between the non-dimensional velocities simplifies to:

$$\bar{U}_\infty = \lambda \sqrt{\beta} \quad (178)$$

Therefore, for a given fluid-structure mass ratio β , our numerical model should predict the onset of static divergence at the theoretical non-dimensional velocity of $\bar{U}_{div} = \sqrt{181\beta}$. The results from our model perfectly match, as can be seen in Fig. 10, where $\bar{U}_{th} = 18.2$, the velocity calculated in [?] matches the nondimensional velocity at which mode 1 becomes unstable.

7.3 Small amplitude vibration

The introduction of a fluid medium fundamentally alters the dynamic response of the bistable plate. As seen in Fig. 11, where the plate is immersed in a completely static fluid ($U_\infty = 0$), all modal frequencies are uniformly shifted

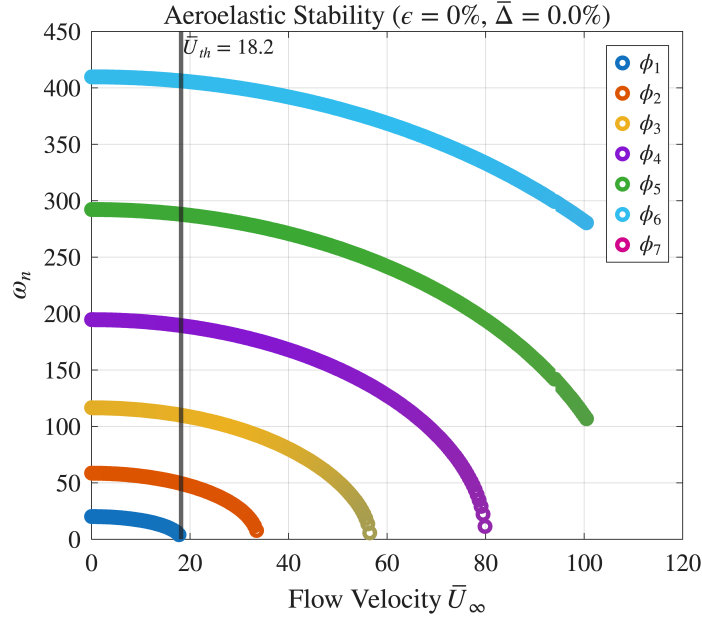


Figure 10: Evolution of the non-dimensional natural frequencies ω_n as a function of the normalized uniform flow velocity $\bar{U}_\infty$ for a flat, unbuckled clamped-clamped plate ($\bar{\Delta} = 0$, $\epsilon = 0$). The uniform flow induces a monotonic decay in the fundamental frequency, leading to static divergence when $\omega_1 = 0$. Using 7 modes for the Galerkin decomposition we found $\bar{U}_{cr} = 18.155$, which aligns very well with the predicted value $\bar{U}_{th} = \sqrt{181\beta} = 18.174$.

downwards compared to their in-vacuo counterparts. This reduction is driven by the added mass of the fluid, encapsulated in the mass matrix M_{kj}^{fluid} . However, when uniform flow is introduced ($U_\infty > 0$), Fig. 12 demonstrates a strong aerodynamic stiffening effect. Unlike the flat plate scenario where flow induces softening, the pre-buckled shape ($\bar{\Delta} > \bar{\Delta}_{cr}$) alters the local flow.

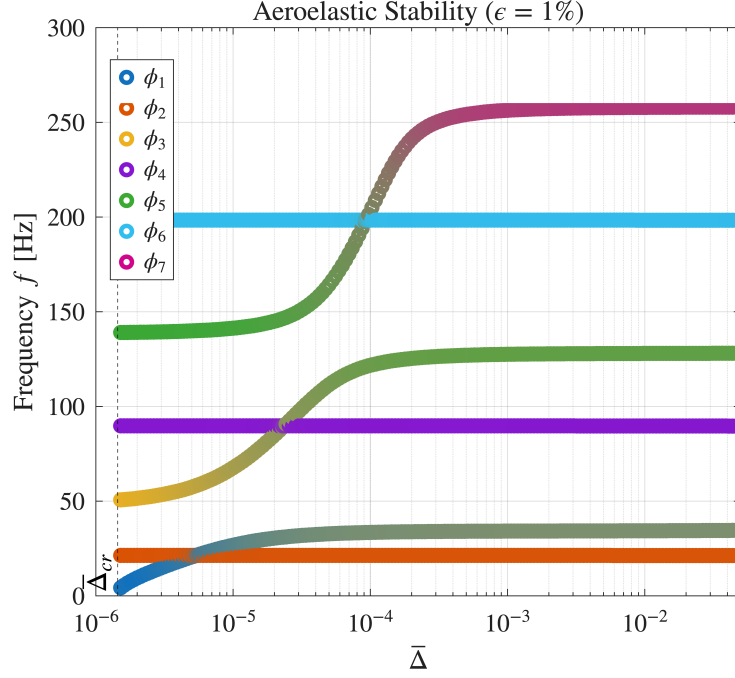


Figure 11: Characteristic frequencies of the different modes for a range of prestrain $\bar{\Delta}$ at 0 flow speed.

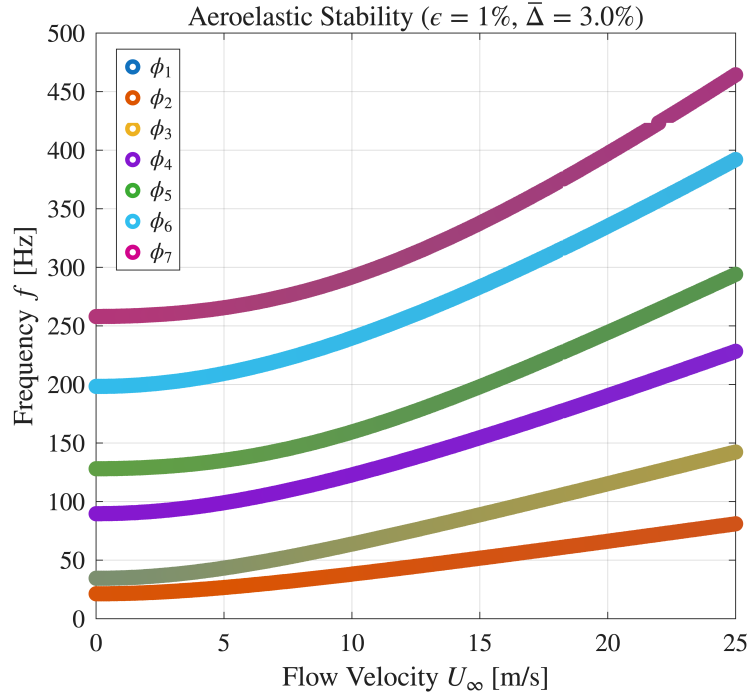


Figure 12: Characteristic frequencies of the different modes for a range of speed, at $\bar{\Delta} = 3\%$. We notice a strong stiffening effect caused by the uniform flow.

8 Beam-KVS interaction results

The parameter maps in Fig. 13 reveal an interesting phenomenon governing the interaction between the plate and the impinging vortices. The vortex shedding frequency generated by the upstream cylinder is fundamentally governed by the Strouhal relation, $f_{shed} = St \cdot U_\infty / D$, scaling linearly with the flow velocity. Simultaneously, the plate's natural frequencies are also increasing due to the aerodynamic stiffening. A scaling analysis shows that for large flow velocities, the plate's mode frequency scales approximately as $f_{mode} \sim \sqrt{K_0 + cU_\infty^2} \sim U_\infty$. Because both the shedding frequency and the structural resonance scale linearly with U_∞ in the high-velocity regime, their ratio f_n/f_{shed} becomes largely insensitive to the flow velocity, depending primarily on the cylinder diameter D . This asymptotic flattening explains the trend of the contour bands in Fig. 13a. Physically, this can represent an advantage: it creates a wide "lock-in" velocity bandwidth where the wake's excitation frequency remains consistently tuned to the plate's resonance, maximizing energy transfer across different flow conditions.

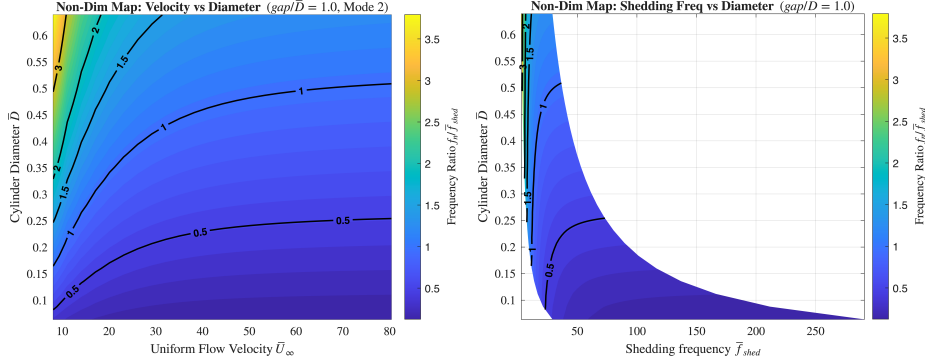


Figure 13: Non-dimensional parameter maps characterizing the fluid-structure interaction between the bistable plate and an upstream Kármán vortex street. (a) Contour map showing the ratio of the structural frequency over shedding frequency (f_n/f_{shed}) across a range of uniform flow velocities $\bar{U}_\infty$ and cylinder diameters $\bar{D}$. (b) The same frequency ratio mapped against the shedding frequency and cylinder diameter.

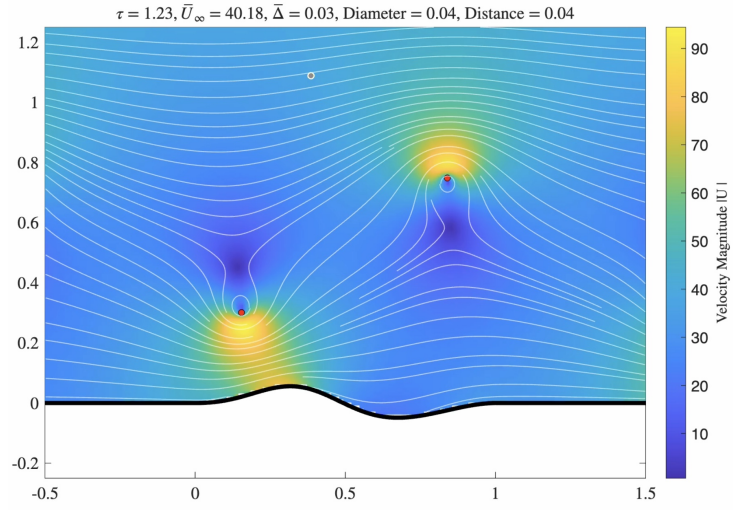


Figure 14: Snap shot in time of the KVS flow interacting with the bistable plate.

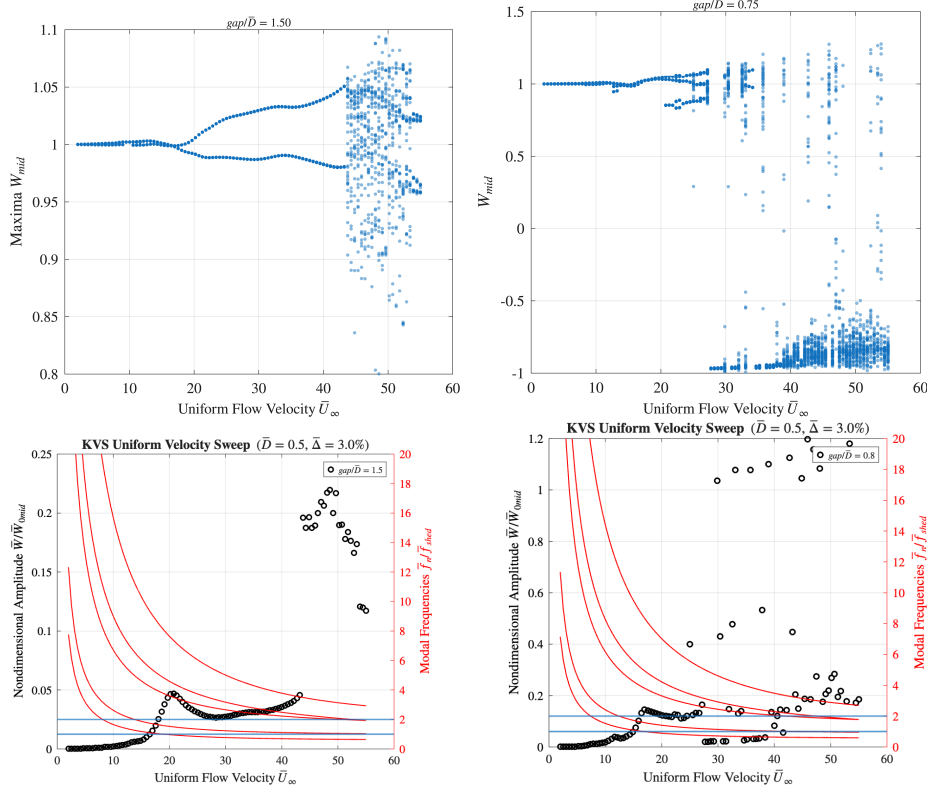


Figure 15: Dynamic response of the plate when excited by a KVS. The shedding frequency changes considerably at low velocities, crossing multiple modal frequency, and settles to a constant f_n/f_{shed} value. This result in multiple peaks in the response amplitude at lower velocities, and a sudden chaotic behavior at higher velocities. Depending on the distance of the shedding vortices form the plate the response change, the onset of chaos happens at lower speed when the gap is smaller, and the frequency peaks show a stronger softening behavior.